# An Overview of the Development of Flexible Sensors

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**Flexible sensors that efficiently detect various stimuli relevant to specific environmental or biological species have been extensively studied due to their great potential for the Internet of Things and wearable electronics applications. The application of flexible and stretchable electronics to device-engineering technologies has enabled the fabrication of slender, lightweight, stretchable, and foldable sensors. Here, recent studies on flexible sensors for biological analytes, ions, light, and pH are outlined. In addition, contemporary studies on device structure, materials, and fabrication methods for flexible sensors are discussed, and a market overview is provided. The conclusion presents challenges and perspectives in this field.**

## 1. Introduction

According to IC Insights' new 2016 O-S-D Report—A Market Analysis and Forecast for Optoelectronics, Sensors/Actuators, and Discretes,[1] in 2015, sales of sensors and actuators achieved a new all-time annual peak of \$10.2 billion. Increasing numbers of sensors and actuators are integrated into new high-volume applications such as the millions of internetworked devices in the Internet of Things and in other wearable systems; therefore, the selling prices of sensors and actuators have declined. The report predicted that the worldwide sales of sensors and actuators would expand by a compound annual growth rate (CAGR) of approximately 6% through 2020 with the help of the Internet of Things. Novel applications in semiautonomous cars, Internet of Things products, and intelligent embedded control were identified as future market growth drivers in the report.[1]

The emerging field of flexible electronics, a technology that is compatible with movable parts and arbitrarily curved surfaces, is promising for a new application paradigm in large-area electronics. Rapid developments in the design of ultrathin sensors and actuators, electronic and optoelectronic devices, and soft biocompatible encapsulating layers are expected to markedly expand the possibilities of flexible electronics from the aforementioned curved panels and foldable displays to soft systems with interfaces that offer curved surfaces and complex geometries.[2–8] **Figure 1** summarizes flexible electronics for the application paradigm in various kinds of electronics.

Conventional sensors, hindered by their rigidity from capturing analytes, suffer from poor-quality signal transduction. Flexible sensors, by contrast, can capture target analytes more efficiently and generate higher quality signals. Production of flexible sensors requires novel approaches in materials design, including active materials and conductors, as well as the selection or synthesis of flexible substrates. Bulk rigid materials become deformable once they are thinned and oriented into nanostructures.[9] For active materials, $\pi$-conjugated low-cost, print-compatible, solution-processable and lightweight organic semiconductors have been used.[3] Non-transition-metal oxides that exhibit high sensitivity and favorable conductivity, such as $ZnO$, $SnO_2$, $In_2O_3$, and $Ga_2O_3$, have also been employed as active materials.[10,11] Among emerging active material candidates for flexible sensors, one-dimensional nanostructure-based materials such as nanowires[12,13] and two-dimensional (2D) nanostructure-based semiconductors including graphene,[14] transition-metal dichalcogenides (TMDs),[15] and black phosphorus[16] all offer distinctive optical and electrical performance characteristics.

Various flexible substrates including polyimide (PI), polyetheretherketone, polyethersulfone (PES), polycarbonate, poly(ethylene napthalate) (PEN), and polyester resins such as polyethylene terephthalate (PET) are often selected for sensing applications. The thermal resistance, chemical resistance, transparency, and flexibility of a material should be considered for practical sensing applications.[17] Conducting materials allowing connection between various components of electronic devices are critical to the functioning of those electronic devices. Nanoparticles, nanotubes, nanowires, and thin films can be employed as core materials for particular processing conditions.[18,19]

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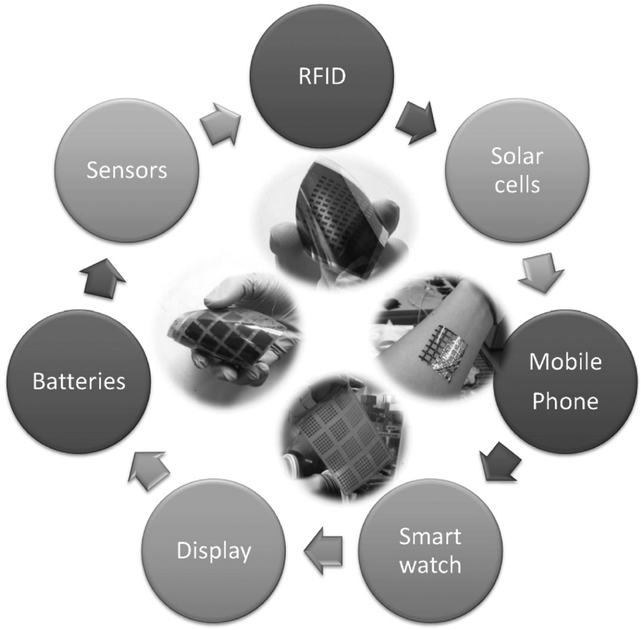


**Figure 1.** Schematic illustration of flexible electronics in development today across a broad range of applications.

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Here, we aim to illustrate various types of flexible sensors. Rather than summarizing all relevant previous work, this paper selects noteworthy work that may suggest crucial future trends of flexible sensors. The discussion presented herein is expected to induce a more fluent exchange of ideas and more intense interest in this emerging field. We summarize the recent state-of-the-art flexible electronics currently employed as flexible light sensors, flexible pH sensors, flexible ion sensors, and flexible biosensors. The selections of materials and the fabrications of devices are included in every part. We also provide a detailed description of engineering technologies with an emphasis on flexible sensor fabrication. We then present market analysis on the world sensor market, printed sensors, and wearable sensors. Finally, a conclusion is given and perspectives for this emerging field are addressed.

## 2. Flexible Sensors for Ultraviolet and Other Photonic-Based Sensing

Light provides energy to the earth and enables our eyes to see the world. Light sensors or photodetectors, which can convert light signals into electrical pulses, are indispensable for most contemporary optical and optoelectronic applications,[20] such as imaging, spectroscopy, fiber-optic communication, and time-gated distance measurements.[21] Fabrication on plastic substrates using large-area mass-production compatible processes such as roll-roll printing facilitates flexible, lightweight, and transparent light sensors.[22] Because of these developments, extensive research interest has been expressed worldwide.

The detection of light typically involves the following four processes: i) light absorption, ii) exciton diffusion, iii) exciton dissociation, iv) charge transfer and collection.[23] A light sensor is required to possess a specific spectral response range, short response time, high quantum efficiency, high responsivity, normalized detectivity, and wide linear dynamic range.[23] It is believed that the sensed spectral range of a photodetector is of vital importance for a myriad of fundamental and practical applications including but not limited to environmental monitoring, communications, and surveillance; thus, this necessitates the development of light sensors working independently in the ultraviolet (UV, <400 nm), visible (400–700 nm) or infrared (IR) (>700 nm) spectral ranges.[23] Fast response is usually required for a light sensor to detect incident light in a short time. The response time is the time required to reach 63.2% of the steady-state or final value,[23] or the curve-rise/decay time constant.[24] Deeper traps are expected to require longer times to release charges, thus resulting in a slow response.[25] Quantum efficiency, the parameter applied to characterize the exciton generation process, is the ratio of electron–hole pairs or photoelectrons generated through light to the number of incident photons. High external quantum efficiency is a prerequisite for high photodetector performance levels.[26] Responsivity (measured in units of A $W^{-1}$), directly reflecting the photodetector sensitivity to light radiation, is defined as the product of generated photocurrent ($J_{ph}$) divided by the incident optical power ($L_{light}$), $R = J_{ph}/L_{light}$.[23] However, the normalized detectivity (in Jones units; 1 Jones = 1 cm $Hz^{1/2}$ $W^{-1}$), $D^* = R/(2qJ_d)^{1/2} = (J_{ph}/L_{light})/(2qJ_d)^{1/2}$, is usually employed as the figure of merit to evaluate the detector sensitivity, where $J_d$


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is the dark current, and $q$ is $1.6 \times 10^{-19}$ Coulombs representing the elementary charge.[21,23] The normalized detectivity is useful because it not only shows how large a photocurrent can be generated by a given light irradiation but also can reflect another vital parameter known as the photocurrent/dark ratio, thereby normalizing for variations in device area and response speed. This allows for comparison between different devices.[21] Not only high photocurrents but also low dark currents are required to achieve high-performance light sensors.[23,27,28] Both large photocurrents and high photocurrent/dark ratios are indispensable when driving and switching displays.[29]

From a structural perspective, two types of light sensors are usually used, namely the photodiode type and the phototransistor type. A photodiode is a semiconductor device with two-terminal electrodes. The simplicity of this structure facilitates fabrication and operation. Photodiodes typically have fast response times, which, in the case of direct-gap semiconductors, can be measured in microseconds or less.[21] Phototransistors, pioneered by William Shockley in 1951, are three-terminal devices; a phototransistor uses a stream of photons as an additional terminal. They are able to realize functions of light detection, photomodulation, electric-field-controlled switching, and magnification of signals in a single device.[23,30] Particularly large gains are achievable through transistor action because the application of a gate bias causes a depletion zone where efficient separation of light-induced excitons into electrons and holes can occur. Subsequently, these charge carriers are accelerated towards source and drain electrodes by a source–drain voltage. The noise of phototransistors can be depressed to a value much lower than the noise values of other light detectors.[23,31] Apart from high photosensitivity and responsivity values, phototransistors are highly suitable for integration into conventional electronic circuits due to the unique structure of field-effect transistors (FETs).[31] Compatibility with CMOS technologies allows for miniaturization. Thus, phototransistors can be applied in Internet of Things devices and comply with Moore's Law.[32,33]

Sensed photons represent some point on a spectrum; narrowband applications such as full-color imaging or visible-blind near-infrared (NIR) detection necessitate spectral (color) discrimination.[34] UV photodetectors have drawn tremendous interest for various scientific and industrial endeavours, including environmental monitoring, space communication, fire alarms, chemical sensing, biological analysis, and military defence.[22,25,35] Visible light sensors also serve noteworthy purposes in daily life. IR detectors exhibit special characteristics with potential for medical, industrial, communication, night-vision imaging, military, and security applications, because IR detectors have the remarkable capability of detecting light transmitted through atmospheric, biological, and other materials.[34,36] Some typical applications in the IR spectral range are telecommunications (1300–1600 nm), thermal imaging (>1500 nm), biological imaging (800–1100 nm), thermal photovoltaics (>1900 nm), and solar cells (800–2000 nm).[37]

Semiconductors can be produced from materials other than silicon to deliver semiconductor bandgaps that are appropriate for specific application requirements.[38] For example, inorganic metal oxides have been widely used for UV photodetectors. Conventional metal-oxide semiconductor photodetectors are based on various metals including tin oxide ($SnO_2$), titanium dioxide ($TiO_2$), zinc oxide (ZnO), indium oxide ($In_2O_3$).

Research on light sensors using ZnO semiconductors is an extremely active subfield of sensor research. ZnO without dopants is an n-type semiconductor. ZnO is attractive in photodetectors because its exciton binding energy is as high as 60 meV, its electron mobility is as high as nearly 100 $cm^2\ V^{-1}\ s^{-1}$, and its remarkably wide bandgap energy has a value of 3.37 eV under ambient conditions.[22,39] Devices constructed with rational structural designs can have high surface-to-volume ratios for harvesting maximal light energy and accelerating efficient exciton dissociation.[40] Bai et al. constructed a UV sensor with a photocurrent/dark ratio as high as $1.2 \times 10^5$ by integrating multiple ZnO nanowires connected in parallel through a contact printing method on a large-scale PET substrate, and they demonstrated the linear scaling of the photoresponse current with the number of nanowires.[39]

Doped ZnO can be a p-type semiconductor. Hsu et al. formed a UV sensor with a La-doped ZnO nanowire on a flexible PI substrate (**Figure 2**).[22] For most n-type metal-oxide semiconductors, humidity is expected to depress the electron transfer

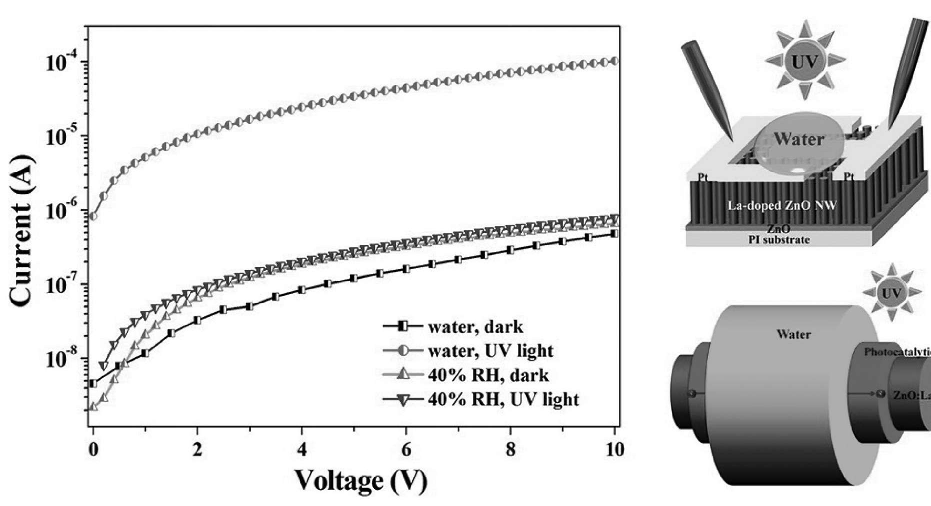


**Figure 2.** *I–V* characteristics of a p-type ZnO:La detector. Reproduced with permission.[22] Copyright 2013, American Chemical Society.

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and the consequent photocurrent responsivity. However, this is not the case for p-type metal-oxide semiconductors. Hsu and his collaborators found that water and humidity were able to improve the photocurrent/dark current ratio up to 212. The authors attributed this performance enhancement to simultaneous dark conductivity decreases and photocurrent enhancements. When under dark conditions, the water replaced the $O_2^-$ ions absorbed on the p-type ZnO nanowire surface, thus reducing the hole concentration and the consequent dark current. However, due to the high photocatalytic activities of the water that covered the ZnO nanowires, more electron and hole exchanges with the water molecules were expected, which dramatically increased the photocurrent by generating a conductive path in the photocatalytic layer.

Two-dimensional graphene has emerged in recent years and has been utilized for a flexible channel material in phototransistors due to its high mobility, which is as high as approximately $10^6$ cm$^{-2}$ V$^{-1}$ s$^{-1}$ at room temperature,[41] along with its high conductivity, favorable chemical stability, and ease of microfabrication and nanofabrication. Graphene's inherent ultrasensitivity presents great opportunities for this material to be explored for new sensors and optoelectronic devices.[42] Lee's group successfully fabricated a flexible UV photodetector composed of ZnO nanorods and graphene on a PI substrate, with a 400-nm-thick poly(vinylpyrrolidone) (PVP) layer as a dielectric sandwiched between bottom and top layers of 20-nm-thick Al2O3. A 60-nm-thick Ni layer was used as the gate electrode (**Figure 3**). This hybrid structure afforded a photocurrent responsivity as high as $2.5 \times 10^6$ A W$^{-1}$, which was also maintained after 10000 cycles of bending. The high light sensitivity was attributed to the efficient electron transfer from ZnO nanorods to graphene, which was confirmed by a negative shift of the Dirac point. The transferred electron concentration per unit of UV intensity was calculated to be as high as $6 \times 10^{12}$ electrons per mW cm$^{-2}$.[41] Lee's group also used a simple spin-coating procedure to fabricate a transparent and flexible array of reduced-graphene-oxide/P(VDF-TrFE) composites as a channel with IR-responsive capability on a PES substrate, which is transparent in visible light but can respond to IR irradiation from human bodies.[43]

Carbon nanotubes have also been employed to improve light sensing. Liu et al.[44] presented a flexible light sensor in the form of a PET sheet that had been coated with single-walled nanotubes (SWNTs) using $Cu_2O$/ZnO hybrid nanofilms. As shown in **Figure 4**, 50-nm ZnO nanofilms and 50-nm $Cu_2O$ layers were sequentially deposited on top of the SWNTs, producing a photodetector with a short response time (<100 ms), excellent repeatability, robust stability, and wide broadband range from 365 to 625 nm without any need to apply an external bias. One noteworthy phenomenon is that once the deposition sequence of ZnO and $Cu_2O$ had been changed, the photocurrent dramatically decreased to one-quarter of that for the $Cu_2O$/ZnO nanofilm photodetector. The authors speculated that whispering-gallery mode resonances could be formed in the $Cu_2O$/ZnO hybrid nanofilms because of the higher refractive index of $Cu_2O$ outside, leading to the enhanced light absorption and trapping.[44]

Sawyer and co-workers reported a hybrid paper device incorporating thorn-like, flexible ZnO-multiwalled carbon nanotubes (MWCNTs); because of its high surface-to-volume ratio and high mobility for efficient carrier transport as well as collection, that paper device exhibited excellent UV sensing.[45] Its maximal photoresponsivity to UV light was 45.1 A W$^{-1}$ at 375 nm, indicating that the external quantum efficiency reached 14 927%. Additionally, the device exhibited fast transient response characteristics with a rise time of 29 ms and a decay time of 33 ms (**Figure 5**).

$SnS_2$ and $MoO_3$ have also been used to fabricate flexible light sensors. For example, Wu and co-workers reported a flexible photodetector using a self-assembled $SnS_2$ microsphere nanosheet on a transparent polypropylene (PP) film to detect light ranging from UV to NIR (300–830 nm).[46] Zheng et al. reported a flexible light sensor with 2D $MoO_3$ single-crystalline belts, with a bandgap of 3.28 eV, on a PET substrate, which exhibited a responsivity of 183 mA W$^{-1}$, a photocurrent/dark ratio of over two orders, and a response time shorter than 1 s.[35]

Quantum dots of materials such as CdSe, CdS,$Fe_3O_4$, CdTe, $TiO_2$, NiO, $Co_3O_4$, and $Mn_3O_4$ have recently attracted particular attention in light sensors. Quantum dots decorate graphene to form compatible heterostructures and achieve size-dependent semiconductor optical properties. Multicarrier effects in nanoparticles are envisioned to provide optical gain and performance enhancement through carrier multiplication.[21] For example, the bandgap of PbS nanoparticles with a diameter of 3 nm or less is three times larger than that of bulk PbS, which is able to generate strong quantum confinement for high-responsivity light sensing in the visible spectral range.[38] Quantum dots can also realize the IR sensitivity that is not accessible by pristine

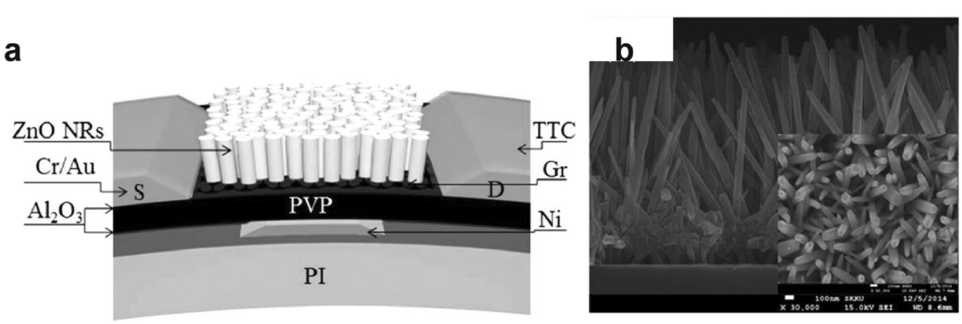


**Figure 3.** a) Schematic illustration of a photodetector composed of ZnO nanorods and graphene on a flexible PI substrate. b) Cross-sectional field emission scanning electron microscope (inset: top-view) image of ZnO nanorods on a flexible PI substrate. Reproduced with permission.[41] Copyright 2015, American Chemical Society.

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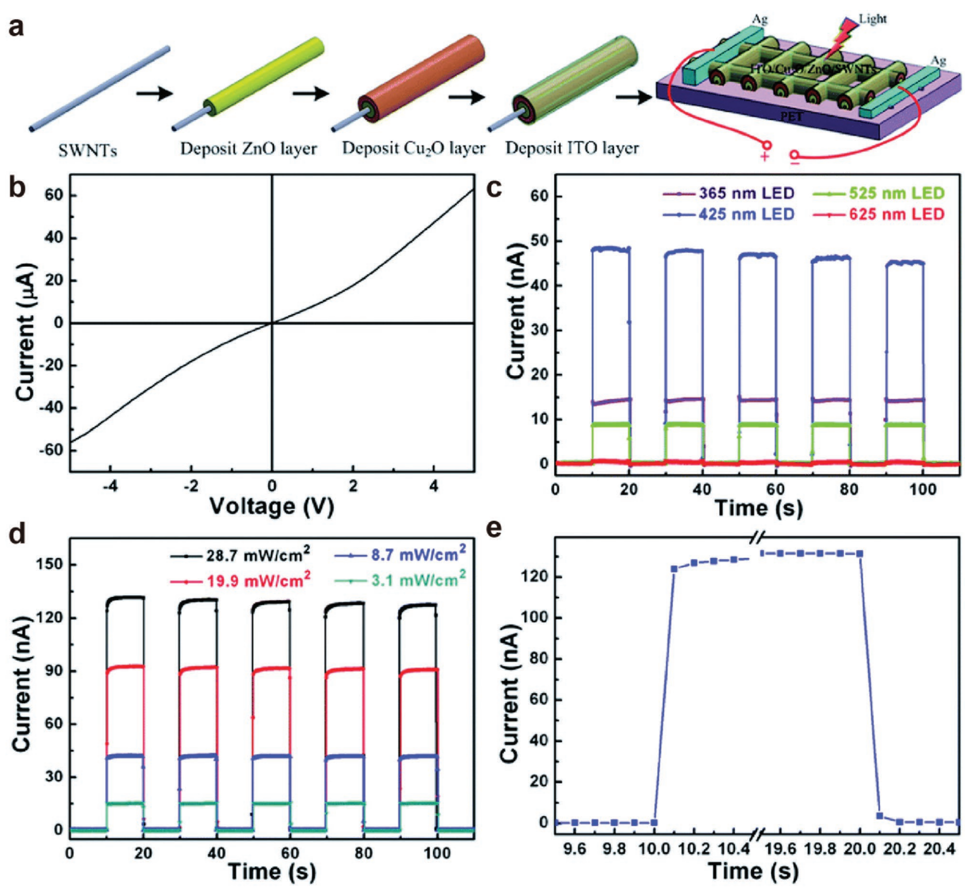


**Figure 4.** a) Scheme to show a photodetector fabrication process involving coating $Cu_2O$/ZnO hybrid nanofilms on a SWNT-coated PET. b) *I–V* curves of the $Cu_2O$/ZnO photodetector with an ITO/$Cu_2O$/ZnO/SWNT structure under dark conditions. c) Time-resolved photoresponse. d) Light-intensity-dependent time-resolved photoresponse. e) Enlarged portion of one photocurrent rising and reset under white light-emitting diode illumination. Reproduced with permission.[44] Copyright 2014, Royal Society of Chemistry.

graphene or ZnO. Liu et al. fabricated a high-performance heterostructure on a flexible PET substrate by integrating synthesized germanium quantum dots, monolayer graphene electrodes, and n-type ZnO, which improved the IR responsivity to approximately 9.7 A $W^{-1}$ at 1400 nm without any sacrifice of the response speed (approximately 40 and 90 µs for rising and decay, respectively). These excellent characteristics along with the high photocurrent/dark ratio of approximately $10^3$ could be guaranteed by the restricted dark current (approximately 1.4 nA, −3 V) due to the effective barrier formed between the graphene and the ZnO interface.[47] Qi and co-workers pioneered symmetric mirror-imaging photoswitching effects in PbS-decorated graphene-based phototransistors; the device responsivities were calculated to be as high as approximately $2.8 \times 10^3$ A $W^{-1}$ at a negative gate bias and approximately $1.7 \times 10^3$ A $W^{-1}$ at a positive gate bias.[42] Quantum dot surface modification is an efficient strategy to improve the photoresponsivity. Sargent and co-workers created films of densely packed colloidal quantum dots with stable benzenedithiol surface passivation, resulting in a 1000-fold decrease of the dark current owing to the elimination of the interaction between primary butylamine ligands and the aluminum contact.[36]

Compared with their inorganic counterparts, organic electronic devices or circuits integrated on lightweight, foldable plastic substrates have numerous fundamental advantages and show great promise for applications in electronic displays, electronic smart cards, or biomedical systems, which are difficult to implement using conventional inorganic electronics.[48] In addition, the manufacturing processes of organic electronics should be much cheaper and more environmentally friendly when fabricated with low-temperature procedures such as a printing technique or roll-to-roll processing.[49–51] The most essential and distinguishing attribute of organic devices is that their chemical versatility allows for easy molecular functionalization and chemical engineering of optoelectronic performance characteristics through artificial molecular designs to afford high sensitivity to optical and electrical stimuli,[52,53] thus allowing for the integration of energy conversion, light detection, and signal magnification in one device.[23] Although conjugated polymers can be easily processed at low cost, with physical flexibility, and at large area coverage,[37] small molecular semiconductors and their composites are more readily available and tend to form more ordered domains for higher charge carrier transport, resulting in greatly improved device reproducibility.[23]

High mobility is critical for light sensors; it determines the photocurrent because the conductivity ($\sigma$) is the product of the number of carriers ($n$) times the mobility ($\mu$) and electronic charge ($e$).[52] Conjugated organic molecules that have strong

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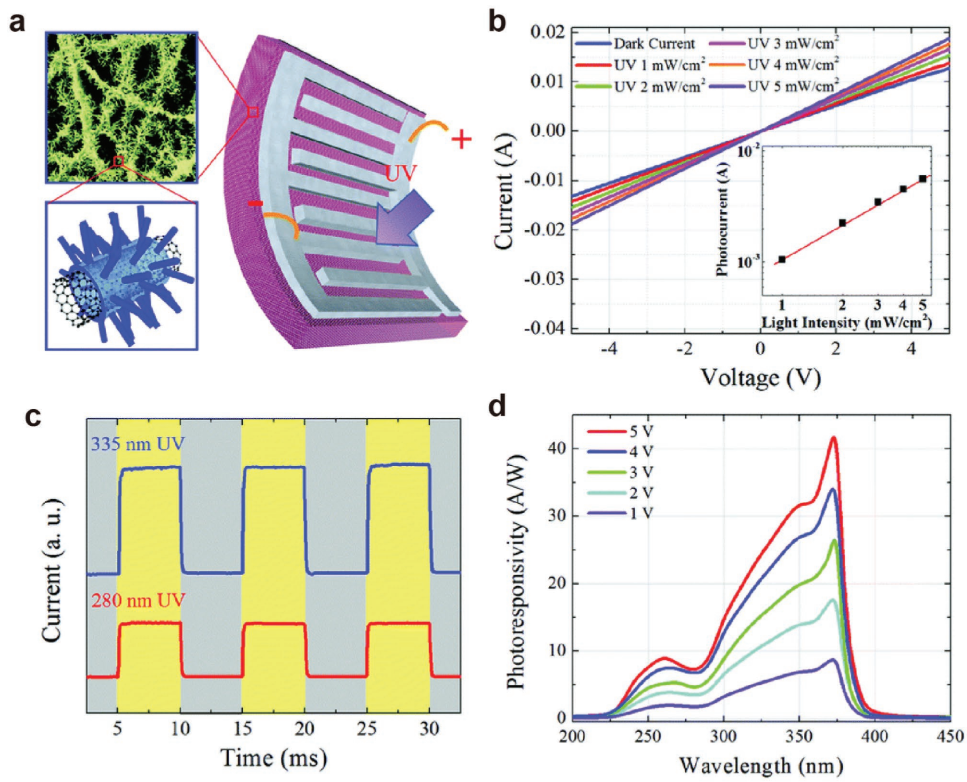


**Figure 5.** a) Scheme to illustrate a ZnO-MWCNT hybrid paper UV photodetector. b) Typical *I–V* curves of the UV photodetector. c) Transient response and d) photoresponsivity spectra of the UV photodetector. Reproduced with permission.[45] Copyright 2014, Royal Society of Chemistry.

intermolecular interactions through $\pi$–$\pi$ overlap stacking are expected to offer high mobility.[53] Li and co-workers fabricated phototransistors using single-crystalline sub-micrometer and nanoscale ribbons of n-type F16CuPc, and they achieved a photocurrent/dark ratio as high as $4.5 \times 10^4$ at a gate bias of –6 V. They also observed that a bias voltage significantly increased the photoresponse, probably because the applied gate bias on the transistors provided an effective method for the separation of the light-induced excitons, benefiting the formation of the transistor conducting channel.[52] As expected, nanocoils of an anthracene derivative in *J*-aggregates that was driven by weak $\pi$–$\pi$ overlap did not show any photoresponse.[53,54] Apart from the crystal morphology, the crystal orientation also significantly influenced the light response.[55]

However, some defects can enhance the photocurrent responsivity. Kim and co-workers employed plasma and ion irradiation on graphene to induce the formation of different types of surface defects such as vacancies, graphene islands, doping, and impurities, which disrupted the honeycomb lattice. These defects strongly scattered the incident light to offer more effective light absorption than pristine graphene, while simultaneously reducing the carrier recombination and facilitating continuous channels for charge carriers, leading to a substantially enhanced photocurrent.[56]

The heterojunction is another critical factor to be considered for organic light sensors. For example, quinacridone thin films prepared through vacuum sublimation only showed responsivity levels lower than 1 mA W$^{-1}$, but a solution-processed film could offer greater responsivity values due to the formation of bulk heterojunctions.[51] Heiss and co-workers prepared a paper photodetector by painting riboflavin-functionalized quinacridone nanocrystals on paper using a paint brush; they sputtered gold through a shadow mask over the nanocrystals to create four electrodes (**Figure 6**a). Although the sample surface was

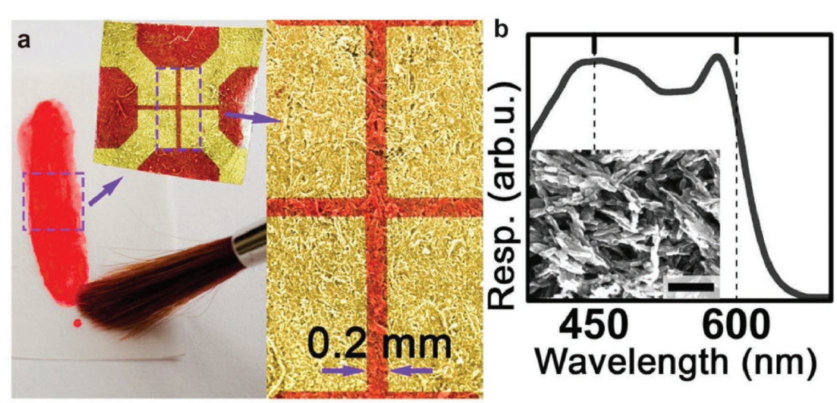


**Figure 6.** Photoconductivity in films of quinacridone micro–nanocrystals with riboflavin myristate ligands. a) Device structure was painted on paper. b) Photoresponsivity spectrum measured on the device. The inset shows the used micro–nanocrystals (scale bar, 200 nm). Reproduced with permission.[51] Copyright 2014, American Chemical Society.

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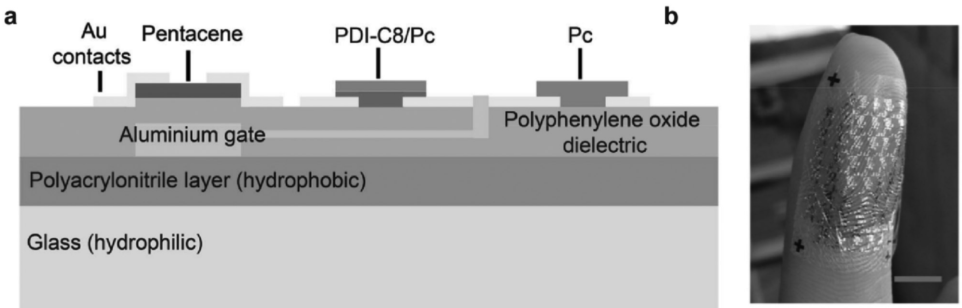


**Figure 7.** a) Cross-sectional diagram of a photosensor consisting of three components: an OFET, OLDR, and OR. b) Flexible sensor arrays on the human finger. Scale bar, 1 cm. Reproduced with permission.[29] Copyright 2015, Wiley-VCH.

substantially rough, they easily measured a photoconductivity spectrum with a favorable signal-to-noise ratio (Figure 6b), and the responsivity reached 1 A $W^{-1}$. The riboflavin-coated quinacridone device was found to exhibit not only a high responsivity but also a high dark resistance of 100 GΩ, leading to a high normalized detectivity of $3 \times 10^{13}$ Jones.[51]

Heterojunctions with different multilayer structures have also been developed for photoresponsivity improvement. The formed multilayer can create blocking electrodes for suppressing a thermally induced dark current, but the bulk heterojunction still allows the transport of photogenerated electrons to opposite electrodes.[28]

Despite extensive research on photodiodes and phototransistors, the photocurrent/dark ratio is still lower than six orders of magnitude.[24] Hu, Liu and co-workers made a breakthrough with organic light-dependent resistors (OLDRs); each was composed of an organic resistor (OR) as a load resistor, with an organic field-effect transistor (OFET) as a readout element. This type of photosensor is of the transconductance type rather than the conductance type; thus, this type can provide a photocurrent/dark ratio up to eight orders of magnitude higher than the other type can (**Figure 7**).[29]

We do not intend to list all relevant work in this miniature review. For clarity, we compare recently reported flexible photosensors in **Table 1**. Because some studies did not report the normalized detectivity, to compare the performance levels of different devices would be challenging. Although apparent progress has been made, further research is required for optimizing performance.

## 3. Flexible Biosensors

OFETs are attracting increasing attention as reliable devices because of their desirable characteristics that include their applicability in organic circuits such as inverters, oscillators,

**Table 1.** Typical flexible light sensors reported recently.

| Channel material | Structure | Substrate | Bias [V] | Wavelength [nm] | Photocurrent/dark ratio | Maximum responsivity [A $W^{-1}$] | Rise time [s] | Decay time [s] | Ref. |
|---|---|---|---|---|---|---|---|---|---|
| ZnO nanorods/graphene | phototransistor | PI | 5 | 365 | 1.2 | $2.5 \times 10^6$ | / | / | [41] |
| La-doped ZnO nanowires | photodiode | PI | 10 | 365 | 212.1 | / | / | / | [22] |
| ZnO nanowires | photodiode | PET | 3 | UV | $1.2 \times 10^5$ | / | / | / | [39] |
| ZnO/Ge quantum dot decorated graphene | phototransistor | PET | −3 | 1400 | $10^3$ | ≈9.7 | $4 \times 10^{-5}$ | $9 \times 10^{-5}$ | [47] |
| ZnO-multiwalled carbon nanotube hybrid paper | photodiode | None | 5 | 375 | / | 45.1 | 0.029 | 0.033 | [45] |
| Graphene-oxide-decorated ZnO nanostructures | photodiode | PDMS | 10 | UV | 116 | / | 6 | 3.5 | [57] |
| Cu2O/ZnO | photodiode | PET | 0 | 365–625 | 520 | / | 0.1 | 0.1 | [44] |
| ZnO tetrapods | photodiode | PET | 20 | 365 | 45 | / | 0.9 | 0.9 | [58] |
| ZnO nanorods | photodiode | PET | 0.5 | 365 | / | / | 100 | 120 | [59] |
| $SnS_2$ nanosheet | photodiode | PP | 5 | 300–830 | <3 | $2.1 \times 10^{-4}$ | ≈0.8 | ≈0.6 | [46] |
| $MoO_3$ nanosheet | photodiode | PET | 10 | 380 | 369 | ≈0.18 | <1 | <1 | [35] |
| InGaZnO | photodiode | PET | 2 | 310 | $3.1 \times 10^3$ | 0.45 | 0.8 | 33.8 | [60] |
| Si | phototransistor | PET | <0.5 | 400–700 nm | $10^5$ | 52 | $5 \times 10^{-5}$ | $1 \times 10^{-4}$ | [20] |
| Riboflavin-functionalized quinacridone nanocrystals | photodiode | paper | 0 | 550 | 190% | 1 | / | / | [51] |
| PDI/Pc | OLDRs | PET | −22 | visible | $10^8$ | / | / | / | [29] |

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and logic circuits without the necessity of advanced patterning techniques.[61,62] Moreover, as the development of industrial printing technology has advanced, the cost of fabricating OFETs on flexible substrates has decreased. Sensors based on OFETs with organic small molecules and polymers have been successfully demonstrated in detecting a wide range of analytes in vapor, ions, and light systems. In addition, biological sensors based on organic electronics are expected to be widely used for detecting various chemical analytes for environmental monitoring, food security, and prophylactic physical examination.[63] In this section, we focus on the recent development of flexible OFET-based biosensors. The device fabrication, operation, and working principles of detecting analytes in various media is explained.

### 3.1. Organic Field-Effect Transistors

OFETs, which can be applied as fundamental components in price tags, smartcards, and sensors, are desirable for the manufacturing of low-cost, large-area devices with plastic substrates. An OFET device contains three different components: a thin semiconductor layer, a gate insulator layer, and three metal electrodes. The semiconductor layer is usually fabricated with small molecules or polymer materials. For small molecules, the semiconductor film is usually fabricated by thermal deposition. Regarding the conjugated polymers, the film is formed by spin-coating or printing methods. Research is still ongoing about how to improve the quality of semiconductor films by tuning fabrication conditions. The insulator layer separates the gate electrode and semiconductor film. For the fabrication of insulator films, polymeric dielectric materials (e.g., poly(4-vinylphenol), poly(methyl methacrylate), and polystyrene) and inorganic oxide materials (such as $SiO_2$ and $Al_2O_3$) are usually used. The role of the insulator film is paramount for achieving fast and low-power devices, as well as high-stability devices, in various complex environments. Electrodes can be composed of metal and conductive materials such as doped conjugated polymers and metallic nanoparticles. The source and drain electrodes inject charges into the semiconductor film for transport. High-work-function metals such as gold, palladium, platinum, and silver are usually chosen. In the operating state of an OFET, the current flows between the source and drain electrodes when voltages are applied on the gate ($V_g$) and drain ($V_d$) electrodes and the voltage on the source ($V_s$) is 0. Charges are injected from the source electrode into the active layer, and the number of charge carriers accumulated in the active layer is determined by the magnitude of potential difference between the gate and source electrodes ($V_{gs}$). The mobility ($\mu$) is used to gauge an OFET's performance; it shows how easily the charges are transported in the active layer under a given electric field. Other device parameters include the on–off ratio (the current ratio of conductive current "on" state to the current closed "off" state), and the threshold voltage (a voltage which turns on the device to allow the current to flow through the active layer). More introductory information about OFETs can be found in a related review.[64–66]

### 3.2. General OTFT Fabrication Approaches

For OFET fabrication, one of the most essential advantages is the adaptability of various fabrication processes to the characteristics of organic materials. Materials' fabrication conditions such as solubility, melting point, and deposition temperature can be directly regulated by structural design and functionality, contributing a wide range of high-performance, low-cost, flexible, and stable organic materials. One method of semiconductor film fabrication is thermal evaporation. Small molecules are evaporated to form a film at a high vapor pressure and high-temperature state. The film quality can be controlled by changing the deposition speed and substrate temperature. Solution spin coating is another procedure commonly used to form a uniform and flat film. The film thickness can be regulated by changing the rotation speed and solution concentration. The volatile solvent can be removed by a postannealing process. Furthermore, the temperature condition is vital to controlling the film quality. However, the aforementioned methods are usually used in the laboratory. In industrial fields, screen printing or ink-jet printing approaches, which are suitable for roll-to-roll processing, are employed.[67–70] They lead to low-cost, large-scale device production on flexible substrates, showing obvious advantages in device commercialization.

### 3.3. Material Considerations for Biosensors

Among various biosensors, DNA is one of most popular biomaterials for detection processes. Lin et al. fabricated label-free DNA sensors by combining organic electrochemical transistors with flexible microfluidic systems.[71] The device performance levels showed little difference at various bending states. In these systems, single-stranded (ss) DNA samples are modified on top of gold gate electrodes as DNA probes to detect complementary DNA targets. The DNA sensor detection limit can be set to 10 pM through applying an electric pulse. Lee and co-workers reported DNA sensors using low-temperature solution-processed flexible In-Zn-O (IZO) TFTs.[72] They employed a dry–wet method to modify DNA nanostructures on the IZO surface, and the sensing limit was estimated to be 0.5 µL of 50 nM DNA target solution. The sensing mechanism for current changes was based on the oxidation of DNA. Serre et al. fabricated novel DNA sensors using a silicon nanowire through a filtration method.[73] Device networks demonstrated favorable reproducibility and homogeneity and were compatible with different types of substrates. Karnaushenko et al. fabricated a flexible diagnostic platform for early detection of avian influenza virus subtype H1N1 DNA sequences, which was based on Si nanowire field-effect transistors (SiNW-FETs).[74] The sensors showed excellent mechanical properties in bending tests down to a 7.5-mm radius after 1000 consecutive bending cycles. The detection limit of the diagnostic platform could reach $40 \times 10^{-12}$ M within 30 min, which would be very suitable for early stage disease diagnosis. Another relevant research field is glucose. Kwak et al. reported a type of flexible glucose sensor with graphene-based ambipolar FETs using FETs as substrates, which could detect glucose levels in a range of approximately 3.3–10.9 mM.[75] The device could perform high-resolution,

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real-time measurement, showing high advantages of low-cost, wearable, and implantable glucose level monitoring applications. Lee et al. reported a type of pH/glucose sensor based on flexible ion-sensitive field-effect transistors (ISFETs) on PET substrates.[76] In the devices, single-walled carbon nanotubes (SWCNTs) and poly(diallyldimethyammonium chloride) are deposited between two patterned metal electrodes. The electronic conductance changes of the SWCNT nanocomposite were measured to characterize the pH levels and then to detect the glucose levels. For flexible substrates, silk protein was used as both a flexible substrate and enzyme fixed material by You et al.[77] Enzymatic biosensors based on graphene FETs showed a detection range of 0.1–10 mM.[77] Minami et al. studied biosensors for lactate detection based on extended-gate type OFETs.[78] In the sensors, two functional layers were modified on a flexible substrate for an enzymatic redox reaction of lactate. The reported detection limit was approximately 66 nM and the quantification limit was 220 nM. The advantage of the extended-gate structure was to avoid the influence of semiconductor films in the aqueous media. Other types of biosensors for detecting trimethylamine,[79] urea,[80] and multisaccharide[81] have also been fabricated and studied.

## 4. Flexible pH Sensors and Ion Sensors

### 4.1. Flexible pH Sensors

pH is a measure of proton activity and is defined as:

$$\mathrm{pH} = -\log a_{\mathrm{H}}^{+} \tag{1}$$

where $a_{\mathrm{H}}^{+}$ defines the proton activity. The strategies for pH sensing can be categorized into electrochemical sensing and spectrometric sensing. This review focuses on some developments in flexible electrochemical pH sensors. Electrochemical pH sensing is achieved through the selective bonding of $H^+$ ion to an immobilized receptor. The bonding induces a charge that is subsequently transduced to a readable electrical signal. Electrochemical pH sensing can be further categorized on the basis of sensor structure into sensitive electrode- and ISFET-based sensing. The most common method of measuring pH is through an ion-sensitive glass electrode. Conventional ISFETs are fabricated on silicon substrates. The obvious inadequacy of these methods is their incompatibility with flexible or wearable technologies. Consequently, considerable research effort has been dedicated to the development of flexible pH sensors over the last 25 years.

#### *4.1.1. Ion-Sensitive Electrodes*

Ion-sensitive electrodes work by measuring the potential of a working electrode (in this context, the working electrode has the sensing layer) against that of a reference electrode. In the early years, the driving force behind research was the motivation to study the in vivo ion concentrations in blood in the coronary arteries.[82,83] To accomplish this, flexible ion sensors were fabricated as sensing layers on PI substrates utilizing hydrogen ionophore embedded in a plasticized PVC membrane. Ultimately, the researchers were successful in finding a correlation between artery occlusion and changes in ion concentrations in the extracellular regions within a pig's heart. However, the plasticized PVC membrane was plagued by problems associated with membrane adhesion, reversibility, and sub-Nernstian sensitivity. One of the standout works during the early years involved the screen printing of a $H^+$-sensitive ruthenium oxide film on a flexible polyester film.[84] These films exhibited superior stability, repeatability, and sensitivity compared with the plasticized PVC membranes. Subsequently, a flexible pH sensor with an iridium oxide sensing film on a PI substrate was developed,[85] and sensors of this type have been applied as pH monitors within a live pig's oesophagus,[86] a rabbit's heart, and a human heart.[87] pH sensors based on carbon nanotubes, with surface-functionalized $H^+$ receptors (PANI/PPy), that are spin coated on flexible plastic substrates show enhanced pH sensitivity that is close to a Nernstian response.[88] Flexible nanostructured oxide films, which do not require surface-functionalized $H^+$ receptors, such as $WO_3$ nanoparticles, also exhibit behavior close to the Nernstian.[89] A highly noteworthy development in terms of engineering and application was reported in which a flexible sensor for the measurement of epidermal pH was constructed in a tattoo format (see **Figure 8**).[90] Another intriguing possibility was revealed in which a flexible piezoelectric self-powering unit was used in measuring the pH across a rigid ZnO microwire.[91] A recent study also presented a unique flexible sensor fabrication procedure in which microporous and nanoporous laser-carbonized patterns were transferred to PDMS substrates and subsequently coated with a $H^+$ receptor (PANI).[92]

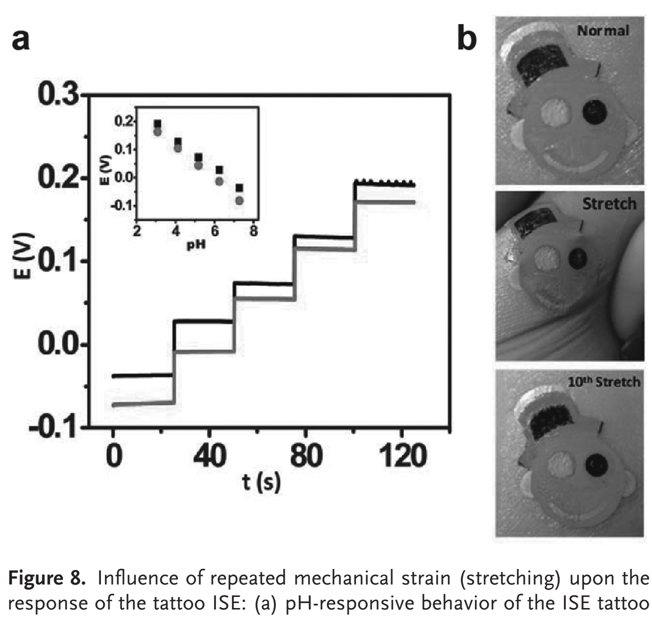


**Figure 8.** Influence of repeated mechanical strain (stretching) upon the response of the tattoo ISE: (a) pH-responsive behavior of the ISE tattoo sensor prior to stretching (black) and following the 40th (red) stretch on GORE-TEX; one unit pH decrement per addition. (b) Images of the tattoo applied to the forearm in the normal state, during stretching, and after the 10th stretch. Reproduced with permission.[90] Copyright 2013, Royal Society of Chemistry.

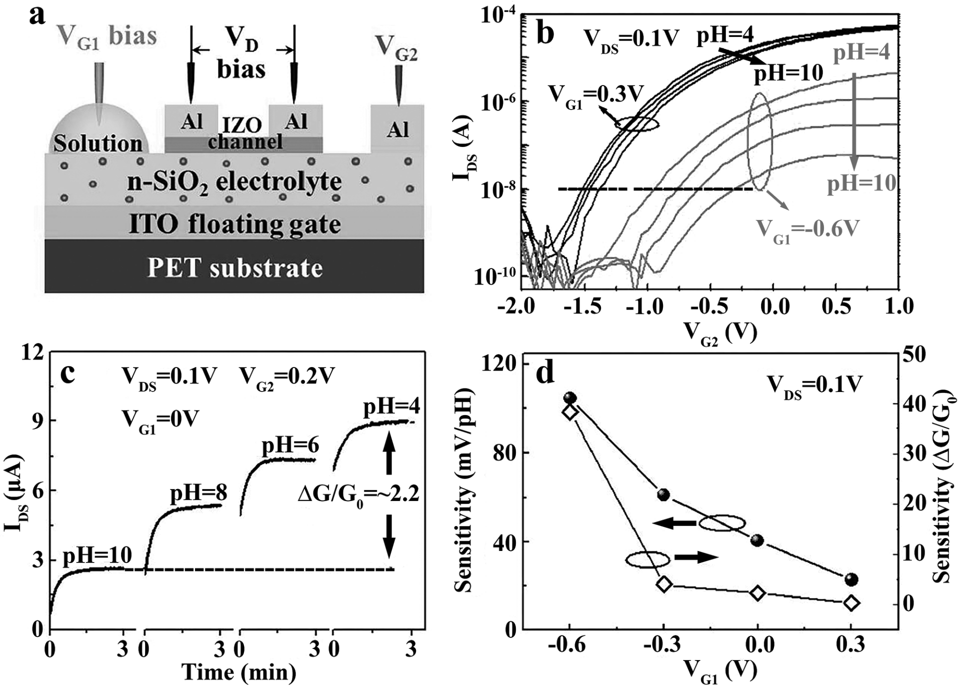


**Figure 9.** pH sensitivities of IZO-based neuromorphic transistor measured in dual-gate synergic modulation mode. Reproduced with permission.[99] Copyright 2015, Macmillan Publishers Limited.

*4.1.2. Ion-Sensitive Field-Effect Transistors*

In ISFETs, the sensing layer can either form the conducting channel (resistance-modulated sensor) or the gate (capacitance-modulated sensor). Flexible resistance-modulated ISFETs using carbon nanotubes[76,93] and a nanostructured oxide film (ZnO nanowall)[94] as the channel sensing layer have been realized. In the study by Maiolo et al.,[94] a Nernstian response was observed. Flexible capacitance-modulated sensors can have their sensing layers deposited directly on top of their dielectric layers[95] or as extended gates. Flexible extended-gate, capacitance-modulated sensors have been demonstrated by making use of pH-sensitive Parylene-C,[96] amorphous IGZO,[97] and roll-roll RF sputtered ITO films[98] deposited on plastic substrates. Notably, in such devices, only the sensing extended gate is flexible, whereas the rest of the device is not. In a recent study, a flexible dual-gate transistor was fabricated in which both gates are capacitatively coupled to a common floating gate.[99] The device structure is shown in **Figure 9**. The advantage of this design is that it can "scale up" the sensitivity of the device to a super-Nernstian response. This device structure is the latest development in flexible ISFETs and holds promise for numerous applications in the biosensing field, where the range of measurement is small.

### 4.2. Flexible Ion Sensor

Ion sensors are functionally similar to pH sensors. Ion sensors are designed to detect specific target ions, usually in aqueous media. The conventional method of selectively detecting ions is through ion-selective electrodes (ISEs) in which glass or glassy carbon electrode tips are covered by an ion-selective membrane. However, from the perspective of wearable technology, they offer little value due to their rigidity, bulkiness, and the requirement of internal filling solutions. Conversely, flexible solid-state ion-selective electrodes (SSISEs) offer attractive properties that can be employed rather trivially in wearable

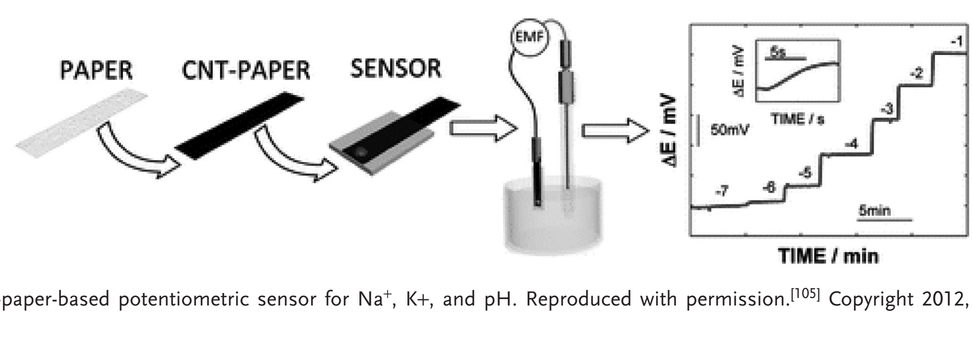


**Figure 10.** Carbon-paper-based potentiometric sensor for $Na^+$, K+, and pH. Reproduced with permission.[105] Copyright 2012, American Chemical Society.

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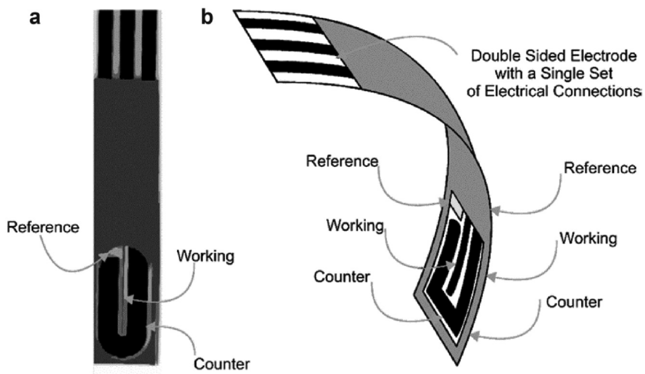


**Figure 11.** Screen-printed electrode (back–back) for Pb(II) anodic stripping voltammetry. Reproduced with permission.[114] Copyright 2015, Royal Society of Chemistry.

technology applications. The general structure of an SSISE is as follows: i) an ion-selective top layer (in contact with the target analyte), ii) an ion–electron transduction middle layer, and iii) a conducting bottom layer (although in some studies, the ion–electron transduction layer and the conducting layer are combined into one layer). SSISEs also have the added benefit of not requiring any internal filling solution (An internal filling solution causes reverse transmembrane fluxes that raise the lower detection limit).[100] This section focuses on the latest developments in flexible SSISEs. In addition, novel flexible ion-sensing platforms are discussed.

The simplest form of an ion sensor is the so-called chemoresistor, in which conductivity of an active material is modulated by its interaction with the target ion.[101] These can be applied on flexible substrates through simple fabrication processes, and this makes them attractive options for low-cost sensing.

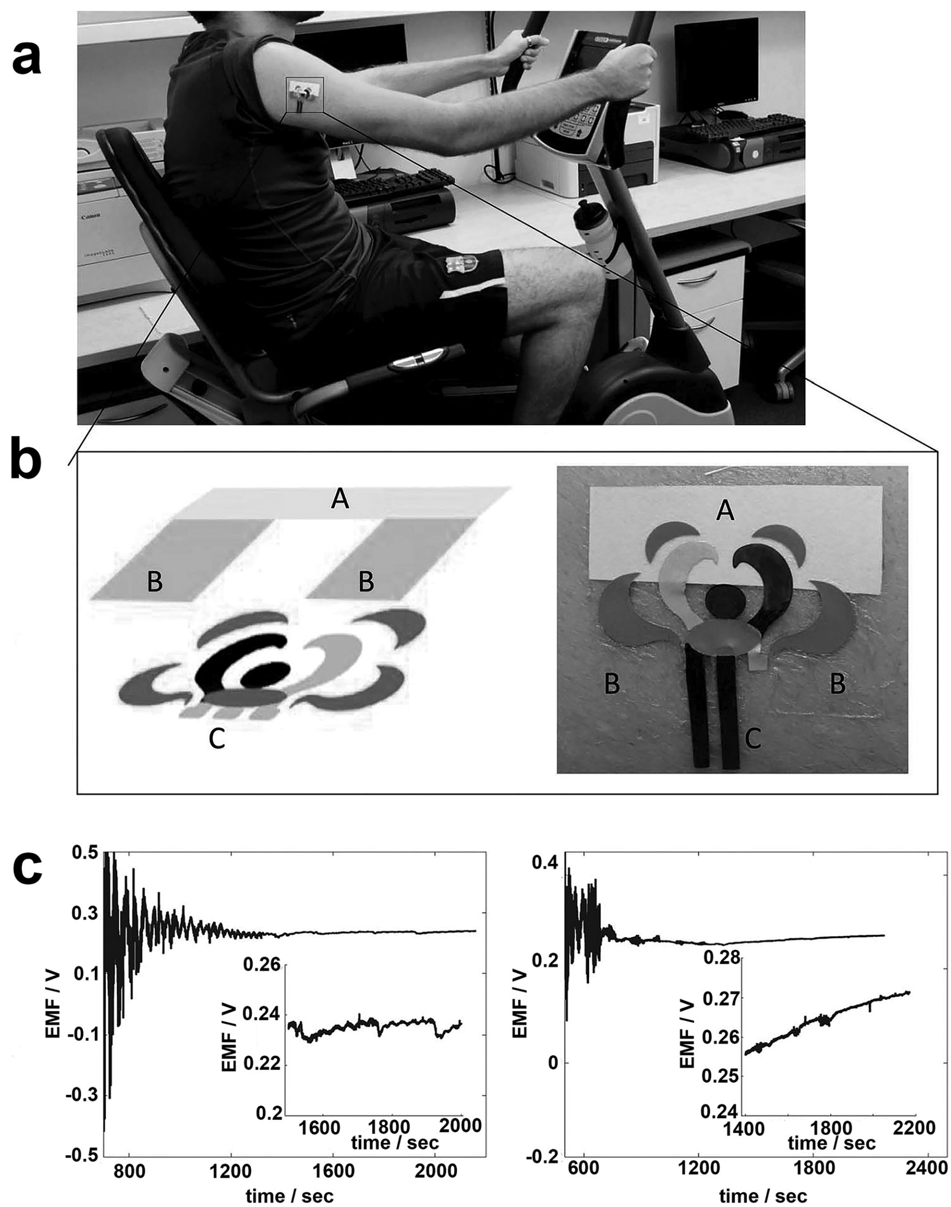


**Figure 12.** Epidermal tattoo sweat sensor for $NH_4^+$ detection. Reproduced with permission.[118] Copyright 2013, Royal Society of Chemistry.

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However, they suffer from several drawbacks such as narrow working ranges, high detection limits, and poor stability. The obvious alternative is an SSISE, for the reasons discussed previously. Most commonly, SSISEs are employed as working electrodes in potentiometric sensors. In a pioneering study, a freestanding strip of a conducting polymer (PEDOT:PSS) was coated with an ion-selective membrane for the selective detection of $K^+$ and $Ca^{2+}$ ions.[102] The performance of these sensors was similar to that of their rigid ISE counterparts. Nevertheless, due to the relatively complex fabrication process and low mechanical strength of freestanding conducting polymer films, research has moved into replacing glassy carbon electrodes with flexible carbon-based substrates for potentiometric SSISEs.

Flexible carbon-based sensors are categorized as carbon–epoxy-resin composite substrates and freestanding carbon-coated cellulose filter papers (filter papers coated with carbon nanotube ink or graphene oxide). $H_2O_2$[103] and Cu(II)[104] potentiometric working electrodes have been demonstrated using carbon–epoxy-resin composite substrates. $K^+$,[105,106] $NH_4^+$,[105] pH (see **Figure 10**),[105,106] $Ca^{2+}$,[106] $NO_2^-$,[107] and $H_2O_2$[108] SSISEs have been built using freestanding carbon paper substrates. The $NO_2^-$ and $H_2O_2$ SSISEs were used in amperometry sensors whereas the rest were used in potentiometric sensors.

Flexible electrodes have also been realized on plastic substrates. Flexible working electrodes on plastic substrates for the detection of sodium dodecylsulfate,[109] $NO_3^-$,[110] $Cl^-$,[111] methyl parathion,[112] $H_2O_2$,[113] and Pb (II)[114] (see **Figure 11**) have all been demonstrated.

The primary motivation driving research into flexible ion sensors and electrodes is to develop epidermal "sweat sensors" for real-time ion monitoring of sweat. Several strategies have been used to develop these sensors: using a nonflexible ISE,[115] flexible electrodes with internal filling solutions,[116] chemoresistors,[117] and tattoo-based sensors (see **Figure 12**).[118,119] A sweat sensor array with sensing capabilities for $Na^+$, $K^+$, glucose, and lactose was fabricated on a PET substrate in a wrist band format (see **Figure 13**).[120] Epidermal sensors designed for ion monitoring in wounds have also been demonstrated (see **Figure 14**).[121–123] This review contains further information on wearable chemical sensors.[124] A developing trend in the field of epidermal sensors is the integration of the sensor with wireless transducers.[119,120,125]

Flexible ion sensors have also been built on FET platforms. A flexible graphene (functionalized with aptamer) FET designed for the detection of $Hg^{2+}$ in mussels was demonstrated with picomolar sensitivity.[126] The emerging field of $MoS_2$ transistors has led to the development of nanomolar sensitive and selective mercury sensors using unfunctionalized few-layer $MoS_2$ structures as the active sensing FET channels.[127] Although this was demonstrated on nonflexible silicon substrates, it could be developed on flexible substrates to produce cheap and sensitive mercury sensors. An offshoot of the FET sensor design is

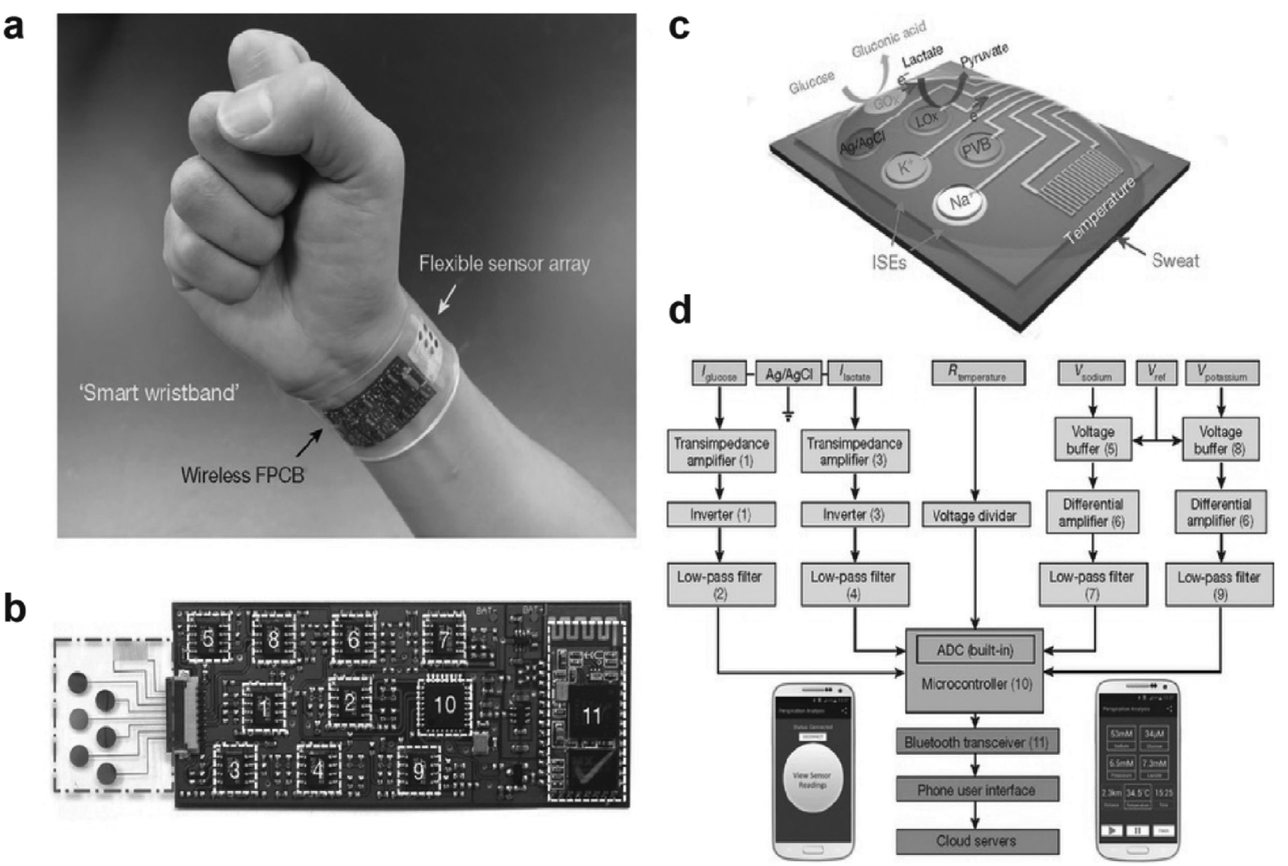


**Figure 13.** Wrist band sweat sensor with integrated wireless transducer. Reproduced with permission.[120] Copyright 2016, Macmillan Publishers Limited.


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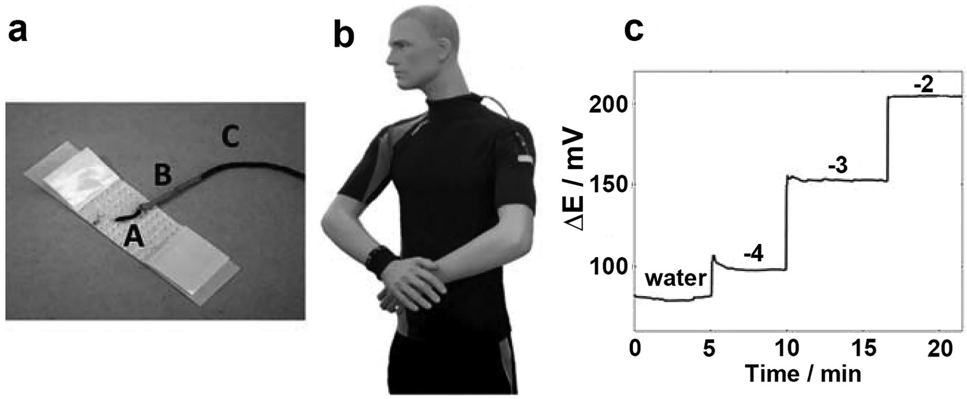


**Figure 14.** Band-aid-integrated $K^+$ cotton/CNT ink sensor. Reproduced with permission.[121] Copyright 2013, Royal Society of Chemistry.

the organic electrochemical junction transistor, which has been used for the detection of $K^+$, $Ca^{2+}$, and $Ag^+$.[128]

Ion exchange across living cell membranes, because of millions of years of evolution, is very selective. The structure of ion channels (which are responsible for ion exchange) has inspired the design of a very appealing ion sensor. Ion-tracked PET substrates are etched to produce nanopores with conical profiles. The walls of these nanopores are then functionalized with selective receptors for specific target ions (see **Figure 15**). These functionalized conical nanopores have been used to detect $Na^+$, $K^+$, and Cr(III).[129] The advantage of these sensors is their low detection limit (nanomolar) and wide linear range.

To achieve general and high-performance multi-analyte discriminant testing, Huang et al. designed and fabricated a multi-stopband photonic crystal microchip based on hydrophilic-hydrophobic patterned substrate. It is remarkable that recognition and analysis of 12 different metal ions and 8-hydroxy-quinoline are realized in one simple sensor (see **Figure 16**).[130]

Real-time measurement of water profiles on flexible substrates using electromagnetic waves is an emerging field. The principle of sensing lies in measuring the shifts in the resonant amplitude and frequency in the microwave to gigahertz range due to the influences of ion concentrations on the dielectric constant of water.[131]

## 5. Approaches to Fabricate Flexible Sensors

In this section, we describe the approaches that are commonly used to fabricate flexible sensors, particularly the different methods used for the preparation of the sensing components, which have been demonstrated previously.[132–151] Here, we list several commonly used methods and discuss the advantages as well as the disadvantages of each approach.

### 5.1. Flexible Sensors Fabricated by Thermal Evaporation

For the preparation of FETs and FET-based sensors, the active layer is usually fabricated by thermal evaporation methods. Pentacene is one of the commonly used thermally evaporated semiconductors for electronics and sensors due

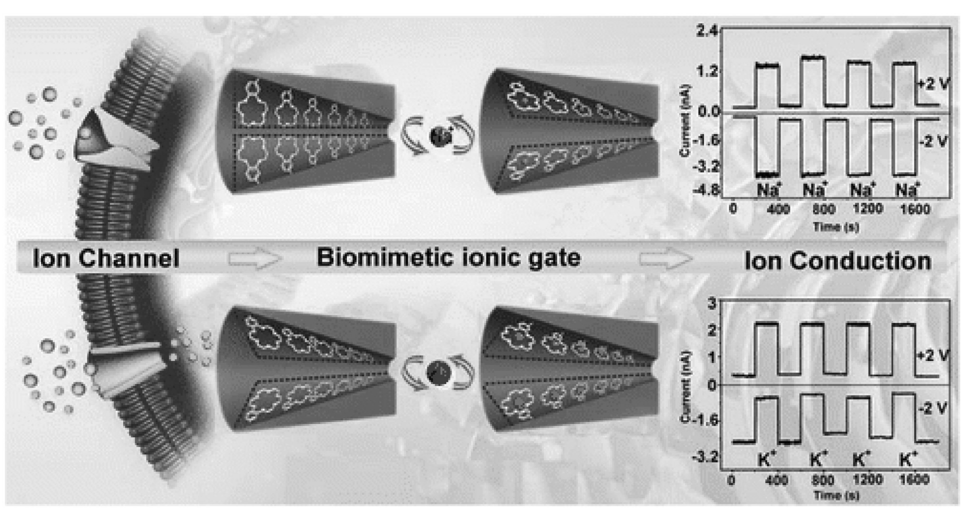


**Figure 15.** Conical nanopore ion channels for the detection of $K^+$ and $Na^+$. Reproduced with permission.[129a] Copyright 2015, American Chemical Society.


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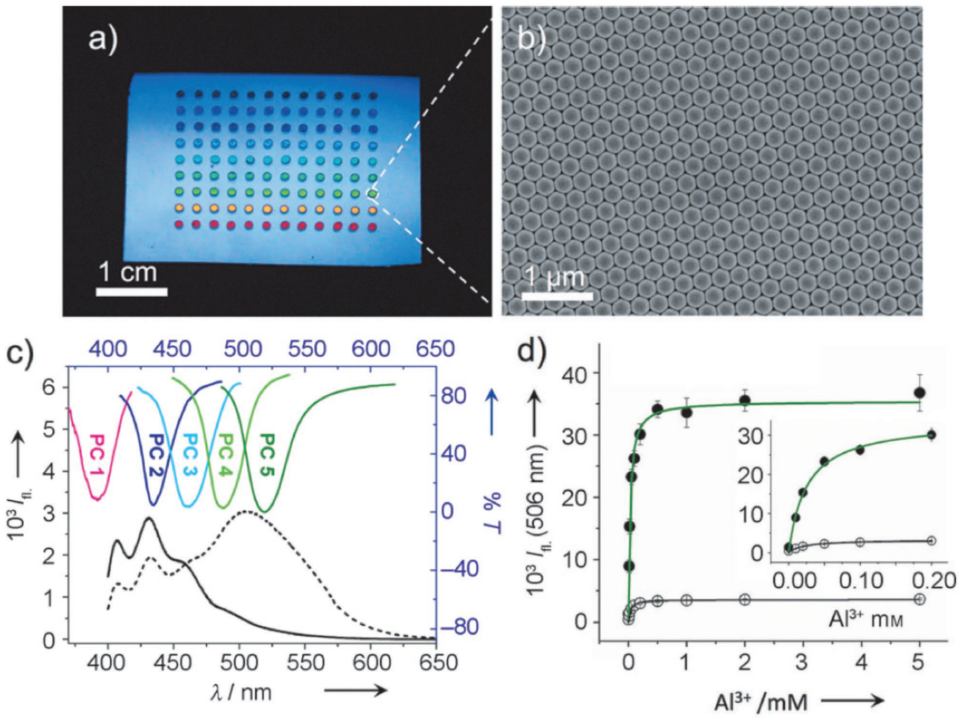


**Figure 16.** a) Photograph of a multi-stopband PC microchip. b) SEM image of the PCs. c) Fluorescence spectra of 8-HQ (c) and 8-HQ-$Al^{3+}$ (a), and transmittance spectra of the PCs. The stopbands of the PCs overlap the different position at three peaks of emission spectra of 8-HQ and 8-HQ-$Al^{3+}$, respectively. d) The $Al^{3+}$ concentration dependence of the fluorescence at 506 nm on PC 5 (*) and on a blank surface. Copyright 2013, Wiley-VCH.

to its relatively high mobility as well as its stability.[152–154] Yakuphanoglu et al. demonstrated a flexible photodetector by employing thermally evaporated pentacene as the sensor's active material.[155] **Figure 17**a shows the device structure, indicating that this device comprises a PES flexible substrate, an ITO gate electrode, a PVP dielectric layer, a thermally evaporated pentacene active layer, and thermally evaporated source and drain electrodes. The sensitivity of this pentacene photodetector was measured with and without UV radiance, and Figure 17b shows the corresponding output curves. As shown in Figure 17b, the flexible pentacene transistor demonstrated very favorable sensitivity for detecting UV light. Sensing films prepared by thermal evaporation are relatively uniform; however, there are also some disadvantages, such as the relatively expensive equipment, low throughput, and high time consumption, all of which render the device unsuitable for industrial production.

### 5.2. Flexible Sensors Fabricated by Chemical Vapor Deposition

In recent years, nanowires, nanorods, and nanotubes have attracted much attention from researchers due to their unique properties for sensors compared with related thin-film structures.[156–160] Nanostructures can be realized by various routes; the chemical vapor deposition (CVD) method is one of the commonly used approaches. Liu et al. demonstrated a CVD-grown ZnO nanowire array on top of flexible polymer substrate, as

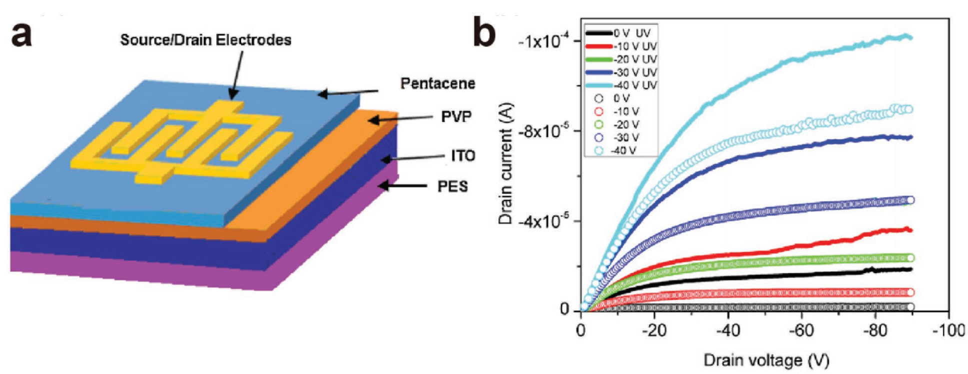


**Figure 17.** a) Schematic of a pentacene photodetector. b) Output curves measured in the dark and under illumination. Reproduced with permission.[155] Copyright 2011, Elsevier B.V.

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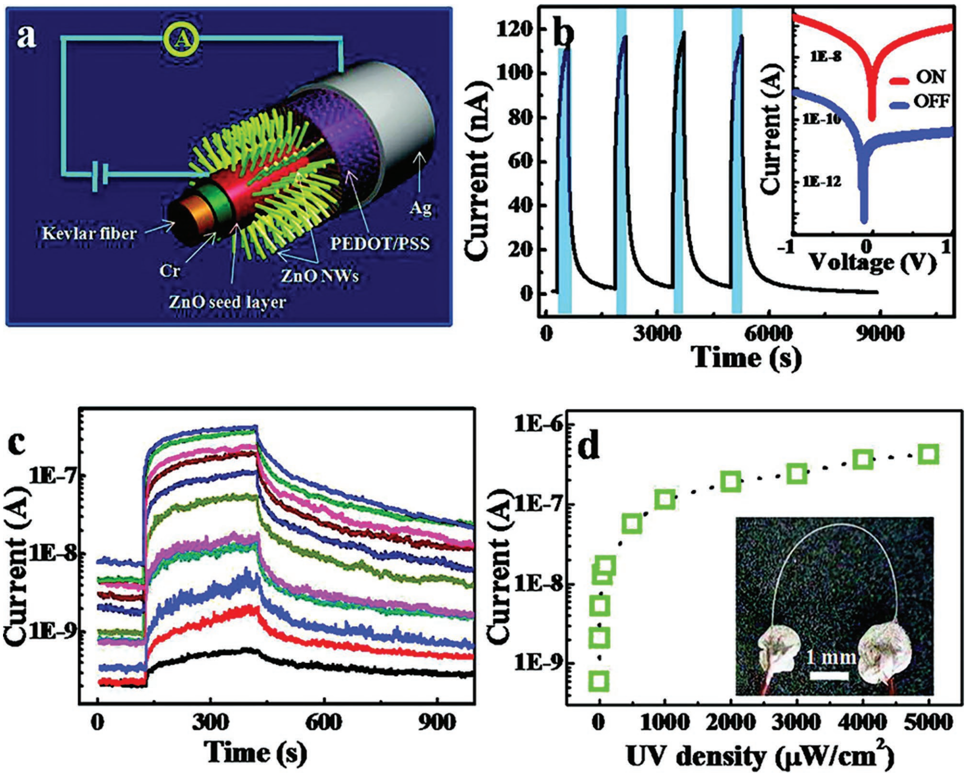


**Figure 18.** a) Schematic diagram of a UV sensor. b) Current curves with UV light on and off. c) Current curves with different UV intensities. d) The curve of response current with increased UV intensity. Reproduced with permission.[161] Copyright 2011, American Chemical Society.

shown in **Figure 18**a.[161] Current curves derived with UV light on and off are shown in Figure 18b, and current curves derived under different UV intensities are shown in Figure 18c. The CVD-grown ZnO nanowire array showed a favorable on–off ratio as well as notable sensitivity.

### 5.3. Flexible Sensors Fabricated by Spin Coating

Spin-coating processes have been widely used in the fabrication of organic electronics and sensors; such processes can reduce fabrication costs greatly compared with CVD

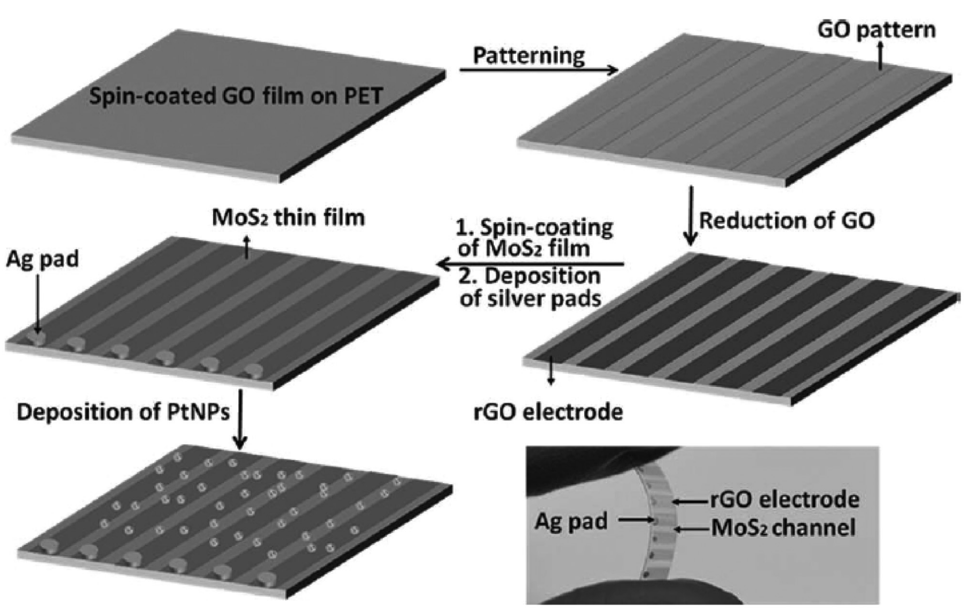


**Figure 19.** Schematic of a spin-coating fabrication process for a flexible gas sensor array. Reproduced with permission.[151] Copyright 2012, Wiley-VCH.

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or thermal evaporation methods. Additionally, spin-coating methods are compatible with low-cost plastic substrates.[162,163] As shown in **Figure 19**, He et al. described a spin-coated single-layer $MoS_2$ gas sensor on a plastic substrate.[151] First, a GO film was patterned on the PET substrate to serve as both source and drain electrodes. Then, a $MoS_2$ active layer was deposited on top by spin coating. The flexible sensor array exhibited high sensitivity to detect toxic gases and its sensitivity was improved further by deposition of PtNPs, as shown in Figure 19.

Zirkl et al. demonstrated optothermal sensors based on spin-coated pyroelectric poly(vinylidenetrifluoroethylene) [P(VDF-TrFE)] copolymer film, as shown in **Figure 20**.[164] The sensing principle is related to the pyroelectric effect in a ferroelectric P(VDF-TrFE) copolymer. These flexible sensors were observed to exhibit favorable sensitivity, as shown in Figure 20d.

### 5.4. Flexible Sensors Fabricated by Printing

Printing technologies have been demonstrated as cost-effective approaches for realizing high-performance electronic components and related sensors based on flexible substrates (PET, PEN, PI, etc.) at proper processing temperatures.[136,165–167] Compared with the other methods, printing routes have some advantages such as simple processing technique, reduced material wastage, and low cost, which make printing methods a very attractive way to fabricate cost-effective sensors. Printed pressure sensors could be wrapped around the arms of robots to sense different pressures.

Contemporary printing approaches could be divided into two categories: noncontact printing and contact printing. With noncontact printing, the patterned structures do not contact with the substrate directly. However, for contact printing, the patterned structures contact with the substrate directly. Compared

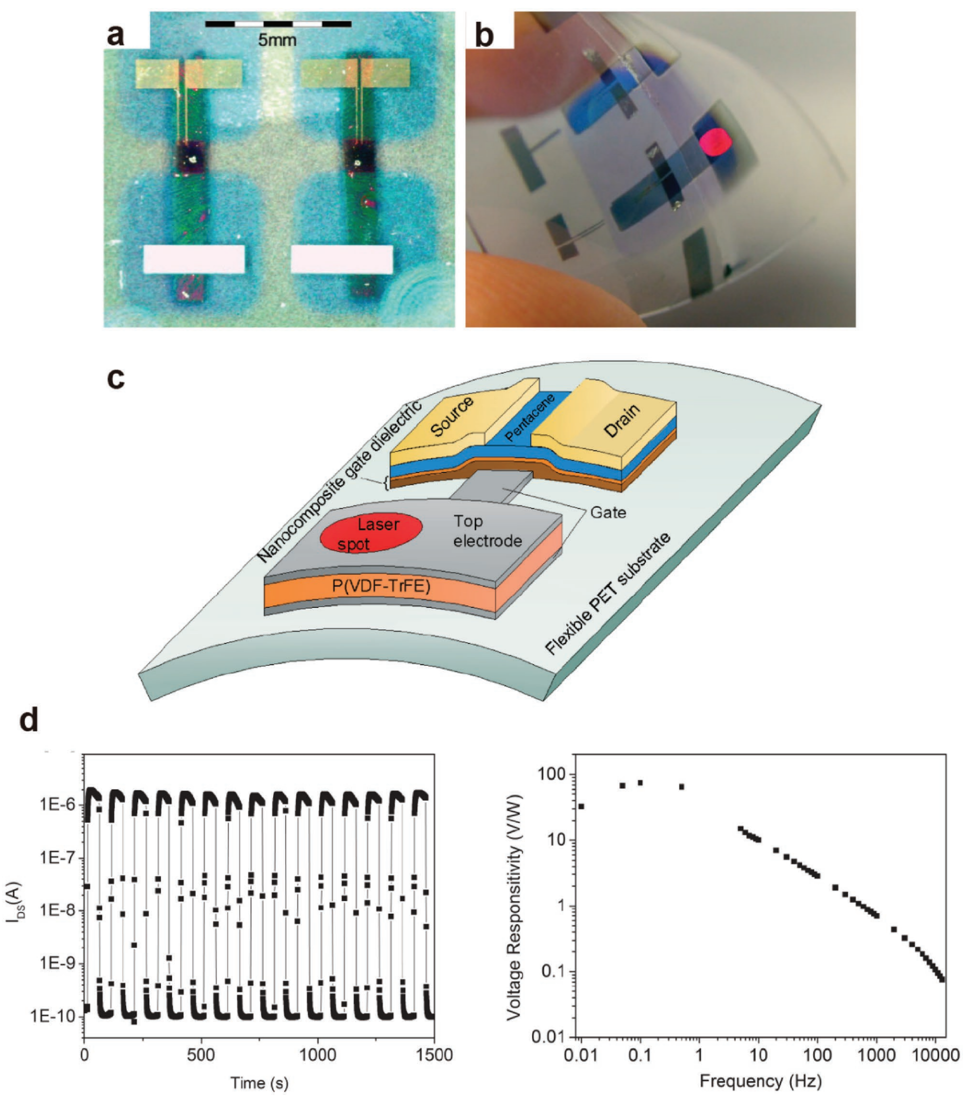


**Figure 20.** a) Images of a flexible optothermal sensor element. b) Schematic of a fully flexible sensor circuit. c) Its operation as a light activated switch. d) On–off cycles and the voltage responsibility versus the frequency for the sensor circuit. Reproduced with permission.[164] Copyright 2007, Wiley-VCH.


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with contact printing,[168] the noncontact approach has the merits of simple fabrication steps, low cost, and reduced material wastage, which makes noncontact methods much more attractive. Here, we mainly discuss the applications of noncontact printing technologies for flexible sensors. Khan et al. described contact printing for flexible electronics and sensors in detail.[167] Among the noncontact approaches, screen printing is the most popular route for flexible sensors. Chang et al. demonstrated flexible large-area pressure sensors by employing screen printing; in their work, the fabrication of the pressure sensors was carried out in two parts.[168] First, holes were drilled and Cu was plated on one side of the substrate followed by 5-µm Au deposition on top of Cu to prevent the oxidation of Cu. Then, a thixotropic material was printed on the other side of the substrate following annealing at 150 °C to form bump structures. For the bottom part, the film fabrication resembled that of the top substrate. A cover layer was then formed on the bottom film using hot pressing to produce post structures. Subsequently, a resistive material was printed on the sensing electrode area. Finally, the two PI substrates were aligned and assembled together to form the pressure sensor.

Another commonly used route to fabricate flexible sensors is ink-jet printing. Ink-jet printing is another solution approach that can form flexible electronics as well as related flexible sensors. As with screen printing, ink-jet printing is cost effective, but some of its drawbacks include low printing speed, poor uniformity of films on the substrates, low pattern resolution, and low throughput due to the low printing speed, which is a challenge for industrial production. Noguchi et al. demonstrated a 33-cm diagonal flexible sensor array on a flexible PI substrate with a dielectric layer fabricated by ink-jet printing.[136] The cross-sectional view of the device structure and the images of the flexible large-area pressure sensor are presented in **Figure 21**a and 21b, respectively. The single pressure sensor and the equivalent circuit diagram are shown in Figure 21c and 21d, respectively.

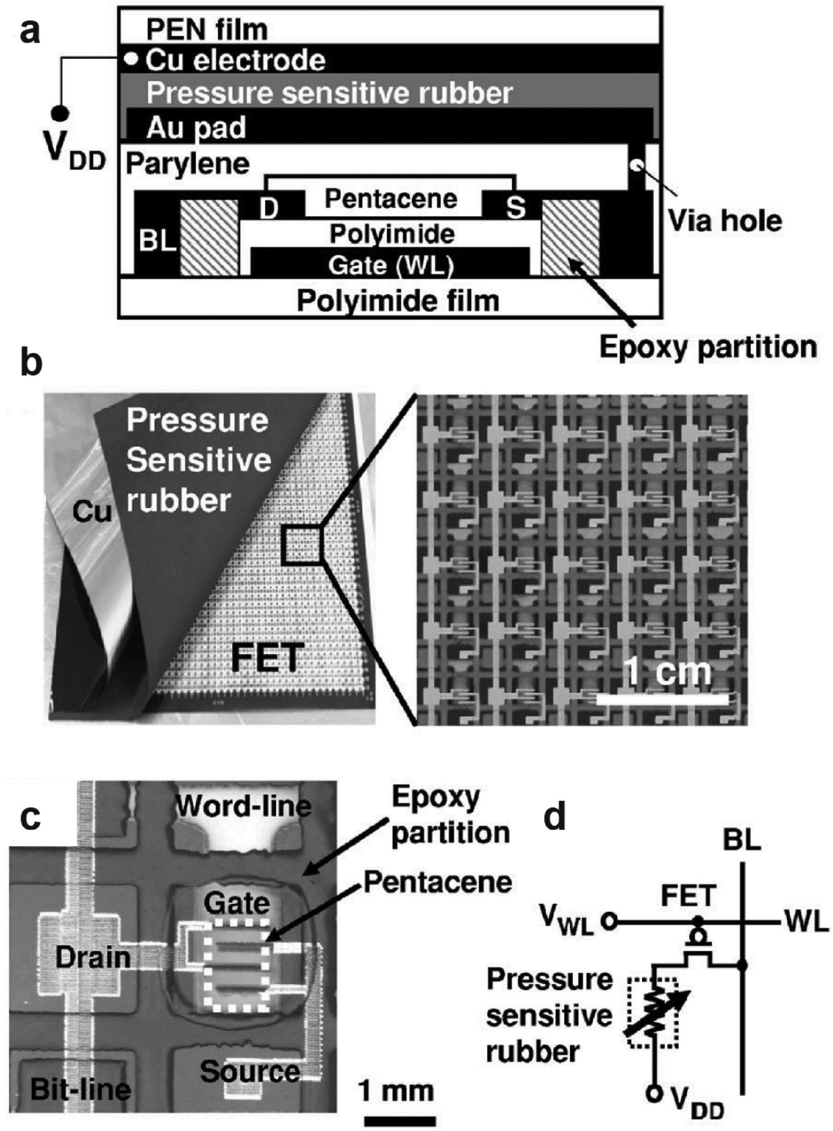


**Figure 21.** a) Cross-sectional view of a flexible pressure sensor; the nanoparticles and the dielectric layer were fabricated by ink-jet printing. b) Microscopy image of the flexible pressure sensor array. c) A single sensor cell. d) Circuit diagram of a pressure cell. Reproduced with permission.[136] Copyright 2006, AIP Publishing LLC.

## 6. Market Analysis

Flexible sensors can be made using inherently flexible materials such as polymer substrates and organic semiconductors. Because a flexible sensor has a high degree of design freedom, it can be trimmed down to different sizes or folded into different shapes. This design parameter is crucial for consumer electronics such as wearable electronics, because wearable electronics must be incorporated into clothing and accessories, which require the sensor to be as tiny as possible. A report from INTECHNO[169] predicted that approximately 2.7% of the sensor market share, approximately €4.97 billion ($5.4 billion), will come from household appliances and consumer electronics, as shown in **Figure 22**.

As mentioned, one major application for flexible sensors in consumer electronics is wearable electronics. A report from IDTechEx predicted that the market for wearable sensors would be worth $5.5 billion by 2025.[170] The market for wearable sensors may seem unremunerative when compared with the market for consumer electronics, because the market for consumer electronics was $5.4 billion in 2016; nevertheless, the market for wearable sensors may only reach that value 10 years later. The cause may be that wearable electronics have


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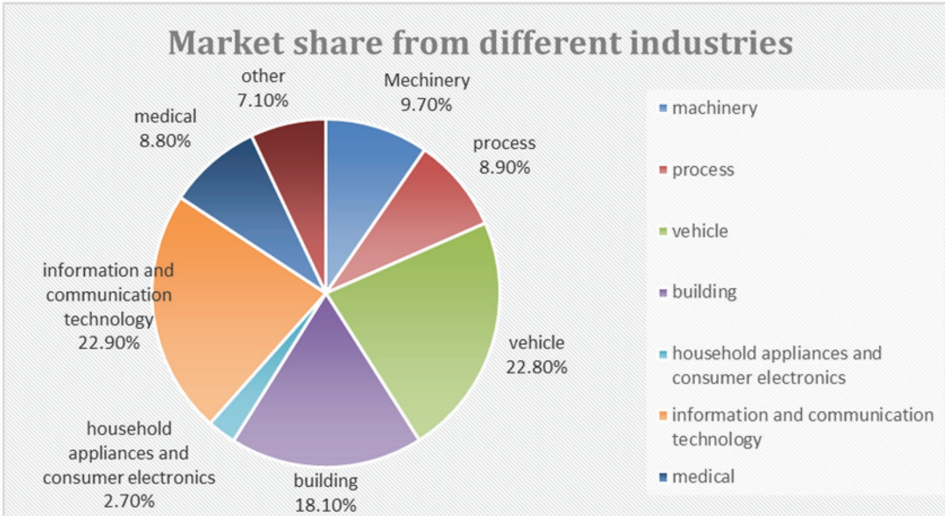


**Figure 22.** Predicted sensor market share of 2016 from different industries. Reproduced with permission.[169] Copyright 2012, INTECHNO.

just emerged recently and are starting from a small base. With proliferating investment from commercial companies such as Google, Microsoft, and Apple, the actual market value of wearable sensors may be higher than previously predicted.

Although one of the major applications for flexible sensors is wearable sensors, flexible sensors can be applied in other areas. Therefore, the market for flexible sensors is expected to be larger than the market for wearable sensors. According to IDTechEx, the market for printed and flexible sensors is expected to reach $8 billion by 2025.[171]

In the following decade, the top five fastest-growing flexible sensor types, according to IDTechEx,[171] are expected to be humidity sensors, temperature sensors, photodetectors, biosensors, and gas sensors, as shown in **Figure 23**. Humidity sensors are expected to have the fastest growth for the next decade because they have started from a low base.[171] If a firm can develop a chemical sensor by detecting different types of gases (gas sensor) or biological molecules (biosensor), that firm might able to catch the trend of a fast-growing market because IDTechEx forecasts that chemical sensors will have the largest wearable sensor market share, as shown in **Figure 24**.[170] Both IDTechEx and INTECHNO claim that the sensor market will be worth billions of dollars. Wearable chemical sensors are expected to have the largest share of the wearable sensor

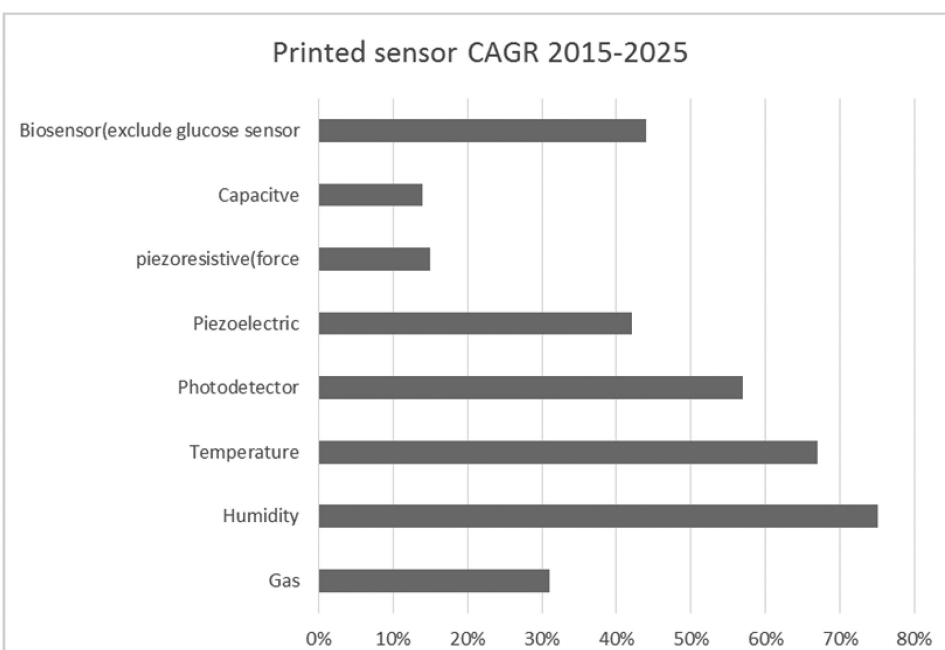


**Figure 23.** Printed sensor CAGR 2015–2025. Reproduced with permission.[171] Copyright 2015, IDTechEx.

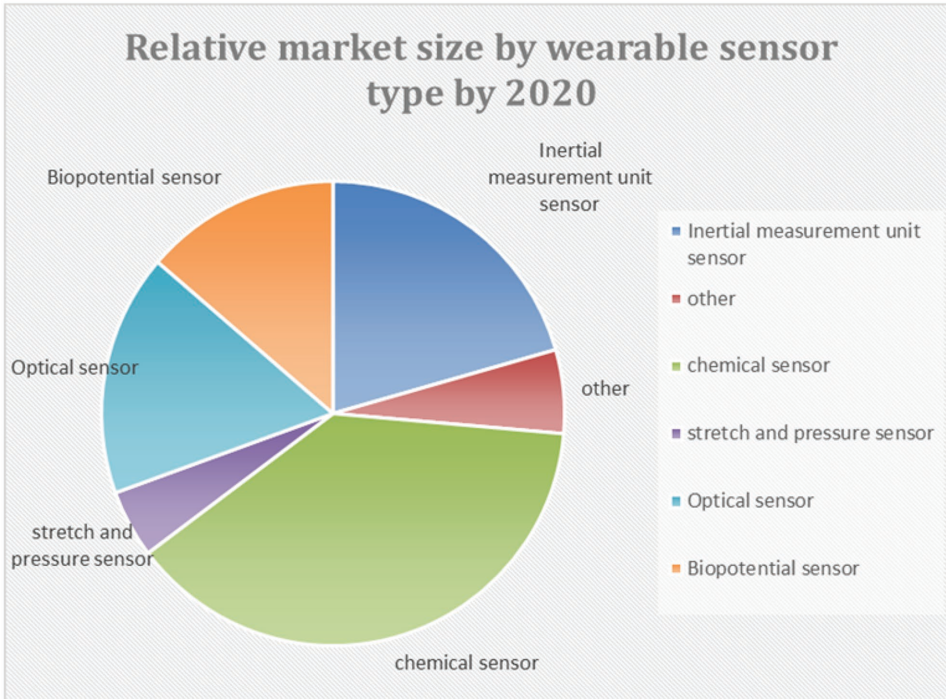


**Figure 24.** Relative market size by wearable sensor type by 2020. Reproduced with permission.[170] Copyright 2015, IDTechEx.

market in the next decade, and humidity sensors are expected to have the highest growth rate for the next decade.

## 7. Conclusion and Perspectives

Considering that most wearable systems, healthcare electronics, and laboratory-on-a-chip testing tools can be expected to come into contact with arbitrarily curved interfaces, the flexibility of sensors is essential for improving their interactions with target systems and improving the reliability and stability of the tests. Hence, flexible sensors hold great promise for various innovative applications in fields such as medicine, healthcare, environment, and biology. Over the past decade, the development of flexible and stretchable sensors for various functions has been accelerated by rapid advances in materials, processing methods, and platforms. For practical applications, new expectations are arising in the pursuit of highly economical, multifunctional, biocompatible flexible sensors.

Numerous opportunities and challenges remain in the future research and development of high-performance flexible sensors. For flexible sensors attached to the human body or its organs, the biocompatibility of the active materials and flexible substrates, including long-term toxicity analyses, is a crucial research area, especially for invasive applications. Innovative utilization of device designs, materials, assembly methods, and surface engineering, as well as interface engineering, can address these challenges. The development of new materials for active layers, substrates, and conductive layers can give rise to soft, stretchable sensors; this emerging paradigm can extend the scope of current technologies for different sensing functions.

Improvement of both flexibility and sensitivity is another challenge for state-of-the-art flexible sensors. The development of novel elastic materials for flexible and stretchable substrates, geometrical electrode designs, combinations of molecular designs in organic materials, and utilization of conceptually novel materials could optimize the trade-off between sensitivity and flexibility.

High-density sensor arrays with multiple functions must be integrated for enabling a high spatiotemporal resolution to realize fully functional flexible electronics. Nevertheless, raising the density of sensors leads to increased crosstalk. Reducing the size of sensors diminishes the amplitude of signals. These problems can be addressed by connecting each sensor with active devices such as transistors to enable local signal amplification and transduction. Designing effective circuits not only facilitates multiplexing but also saves substantial amounts of power. We believe that these challenges can be principally overcome with the aforementioned methods.

The development of sensors is the enabling technology for Internet of Things. A surge in Internet of Things provides plentiful opportunities for spreading out of flexible sensors with reconfigurable shape and size. Thanks to their light weight, thinness, and robustness, flexible sensors can be seamlessly integrated onto any surface to provide users more improved avenues, which is difficult to realize in conventional electromechanical sensors. With the development of polymers, oxides, printing technologies, and CMOS technologies, flexible sensors will unlock a completely novel set of Internet of Things products. The achievement of these fantastic sensing applications will bring us closer to the new electronic era promised by flexible sensors.


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## Conflict of Interest

The authors declare no conflict of interest.


## Keywords

biosensors, flexible electronics, pH and ion sensors, sensors, UV and light sensors

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[1] O-S-D Report, http://www.icinsights.com/services/osd-report/report-contents/, October 2016.
[2] J. Ge, L. Sun, F.-R. Zhang, Y. Zhang, L.-A. Shi, H.-Y. Zhao, H.-W. Zhu, H.-L. Jiang, S.-H. Yu, *Adv. Mater.* **2016**, *28*, 722.
[3] C. Liao, M. Zhang, M. Y. Yao, T. Hua, L. Li, F. Yan, *Adv. Mater.* **2015**, *27*, 7493.
[4] M. Park, Y. J. Park, X. Chen, Y.-K. Park, M.-S. Kim, J.-H. Ahn, *Adv. Mater.* **2016**, *28*, 2556.
[5] Y. Zang, D. Huang, C.-a. Di, D. Zhu, *Adv. Mater.* **2016**, *28*, 4549.
[6] A. Chortos, J. Liu, Z. Bao, *Nat. Mater.* **2016**, *15*, 937.
[7] K. Chen, W. Gao, S. Emaminejad, D. Kiriya, H. Ota, H. Y. Y. Nyein, K. Takei, A. Javey, *Adv. Mater.* **2016**, *28*, 4397.
[8] S.-W. Hwang, C. H. Lee, H. Cheng, J.-W. Jeong, S.-K. Kang, J.-H. Kim, J. Shin, J. Yang, Z. Liu, G. A. Ameer, Y. Huang, J. A. Rogers, *Nano Lett.* **2015**, *15*, 2801.
[9] S. Choi, H. Lee, R. Ghaffari, T. Hyeon, D.-H. Kim, *Adv. Mater.* **2016**, *28*, 4203.
[10] N. Barsan, D. Koziej, U. Weimar, *Sens. Actuators, B* **2007**, *121*, 18.
[11] H. Hosono, in *Handbook of Transparent Conductors*, (Ed. S. D. Ginley), Springer, Boston, MA, USA **2011**, pp. 459–487.
[12] A. Tao, F. Kim, C. Hess, J. Goldberger, R. He, Y. Sun, Y. Xia, P. Yang, *Nano Lett.* **2003**, *3*, 1229.
[13] Z. Fan, J. C. Ho, T. Takahashi, R. Yerushalmi, K. Takei, A. C. Ford, Y.-L. Chueh, A. Javey, *Adv. Mater.* **2009**, *21*, 3730.
[14] O. C. Compton, S. T. Nguyen, *Small* **2010**, *6*, 711.
[15] J. N. Coleman, M. Lotya, A. O'Neill, S. D. Bergin, P. J. King, U. Khan, K. Young, A. Gaucher, S. De, R. J. Smith, I. V. Shvets, S. K. Arora, G. Stanton, H.-Y. Kim, K. Lee, G. T. Kim, G. S. Duesberg, T. Hallam, J. J. Boland, J. J. Wang, J. F. Donegan, J. C. Grunlan, G. Moriarty, A. Shmeliov, R. J. Nicholls, J. M. Perkins, E. M. Grieveson, K. Theuwissen, D. W. McComb, P. D. Nellist, V. Nicolosi, *Science* **2011**, *331*, 568.
[16] X. Zhang, H. Xie, Z. Liu, C. Tan, Z. Luo, H. Li, J. Lin, L. Sun, W. Chen, Z. Xu, L. Xie, W. Huang, H. Zhang, *Angew. Chem.* **2015**, *127*, 3724.
[17] Y. S. Rim, S.-H. Bae, H. Chen, N. De Marco, Y. Yang, *Adv. Mater.* **2016**, *28*, 4415.
[18] C.-H. Chung, T.-B. Song, B. Bob, R. Zhu, Y. Yang, *Nano Res.* **2012**, *5*, 805.
[19] D. Zhang, K. Ryu, X. Liu, E. Polikarpov, J. Ly, M. E. Tompson, C. Zhou, *Nano Lett.* **2006**, *6*, 1880.
[20] J.-H. Seo, K. Zhang, M. Kim, D. Zhao, H. Yang, W. Zhou, Z. Ma, *Adv. Opt. Mater.* **2016**, *4*, 120.
[21] G. Konstantatos, E. H. Sargent, *Nat. Nanotechnol.* **2010**, *5*, 391.
[22] C.-L. Hsu, H.-H. Li, T.-J. Hsueh, *ACS Appl. Mater. Interfaces* **2013**, *5*, 11142.
[23] H. Dong, H. Zhu, Q. Meng, X. Gong, W. Hu, *Chem. Soc. Rev.* **2012**, *41*, 1754.
[24] C. S. Smithson, Y. Wu, T. Wigglesworth, S. Zhu, *Adv. Mater.* **2015**, *27*, 228.
[25] F. W. Guo, B. Yang, Y. B. Yuan, Z. G. Xiao, Q. F. Dong, Y. Bi, J. S. Huang, *Nat. Nanotechnol.* **2012**, *7*, 798.
[26] E. Saracco, B. Bouthinon, J.-M. Verilhac, C. Celle, N. Chevalier, D. Mariolle, O. Dhez, J.-P. Simonato, *Adv. Mater.* **2013**, *25*, 6534.
[27] G. Konstantatos, I. Howard, A. Fischer, S. Hoogland, J. Clifford, E. Klem, L. Levina, E. H. Sargent, *Nature* **2006**, *442*, 180.
[28] X. Gong, M. Tong, Y. Xia, W. Cai, J. S. Moon, Y. Cao, G. Yu, C.-L. Shieh, B. Nilsson, A. J. Heeger, *Science* **2009**, *325*, 1665.
[29] H. Wang, H. Liu, Q. Zhao, C. Cheng, W. Hu, Y. Liu, *Adv. Mater.* **2015**, *28*, 624.
[30] H. Zhang, J. Hui, H. Chen, J. Chen, W. Xu, Z. Shuai, D. Zhu, X. Guo, *Adv. Electron. Mater.* **2015**, *1*, 1500159.
[31] Y. L. Guo, G. Yu, Y. Q. Liu, *Adv. Mater.* **2010**, *22*, 4427.
[32] J. V. Crivello, E. Reichmanis, *Chem. Mater.* **2014**, *26*, 533.
[33] J. Bardeen, W. H. Brattain, *Phys. Rev.* **1948**, *74*, 230.
[34] A. Armin, R. D. Jansen-van Vuuren, N. Kopidakis, P. L. Burn, P. Meredith, *Nat. Commun.* **2015**, *6*, 6343.
[35] Q. Zheng, J. Huang, S. Cao, H. Gao, *J. Mater. Chem. C* **2015**, *3*, 7469.
[36] J. P. Clifford, G. Konstantatos, K. W. Johnston, S. Hoogland, L. Levina, E. H. Sargent, *Nat. Nanotechnol.* **2009**, *4*, 40.
[37] S. A. McDonald, G. Konstantatos, S. G. Zhang, P. W. Cyr, E. J. D. Klem, L. Levina, E. H. Sargent, *Nat. Mater.* **2005**, *4*, 138.
[38] G. Konstantatos, J. Clifford, L. Levina, E. H. Sargent, *Nat. Photonics* **2007**, *1*, 531.
[39] S. Bai, W. Wu, Y. Qin, N. Cui, D. J. Bayerl, X. Wang, *Adv. Funct. Mater.* **2011**, *21*, 4464.
[40] N. Nasiri, R. Bo, F. Wang, L. Fu, A. Tricoli, *Adv. Mater.* **2015**, *27*, 4336.
[41] D. Vinh Quang, T. Tran Quang, D. Le Thai, B.-Y. Kim, S. Siddiqui, W. Lee, N.-E. Lee, *ACS Appl. Mater. Interfaces* **2015**, *7*, 11032.
[42] D. Zhang, L. Gan, Y. Cao, Q. Wang, L. Qi, X. Guo, *Adv. Mater.* **2012**, *24*, 2715.
[43] T. Tran Quang, S. Ramasundaram, N.-E. Lee, *Adv. Funct. Mater.* **2015**, *25*, 1745.
[44] X. Liu, H. Du, P. Wang, T.-T. Lim, X. W. Sun, *J. Mater. Chem. C* **2014**, *2*, 9536.
[45] D. Shao, H. Sun, J. Gao, G. Xin, M. A. Aguilar, T. Yao, N. Koratkar, J. Lian, S. Sawyer, *Nanoscale* **2014**, *6*, 13630.
[46] Y. Tao, X. Wu, W. Wang, J. Wang, *J. Mater. Chem. C* **2015**, *3*, 1347.
[47] X. Liu, X. Ji, M. Liu, N. Liu, Z. Tao, Q. Dai, L. Wei, C. Li, X. Zhang, B. Wang, *ACS Appl. Mater. Interfaces* **2015**, *7*, 2452.
[48] Y. Wen, Y. Liu, Y. Guo, G. Yu, W. Hu, *Chem. Rev.* **2011**, *111*, 3358.
[49] T. Someya, Y. Kato, S. Iba, Y. Noguchi, T. Sekitani, H. Kawaguchi, T. Sakurai, *IEEE Trans. Electron Devices* **2005**, *52*, 2502.
[50] H. Lee, S. Nam, H. Kwon, S. Lee, J. Kim, W. Lee, C. Lee, J. Jeong, H. Kim, T. J. Shin, Y. Kim, *J. Mater. Chem. C* **2015**, *3*, 1513.
[51] M. Sytnyk, E. D. Glowacki, S. Yakunin, G. Voss, W. Schofberger, D. Kriegner, J. Stangl, R. Trotta, C. Gollner, S. Tollabimazraehno, G. Romanazzi, Z. Bozkurt, M. Havlicek, N. S. Sariciftci, W. Heiss, *J. Am. Chem. Soc.* **2014**, *136*, 16522.


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[52] Q. Tang, L. Li, Y. Song, Y. Liu, H. Li, W. Xu, Y. Liu, W. Hu, D. Zhu, *Adv. Mater.* **2007**, *19*, 2624.
[53] R. Li, W. Hu, Y. Liu, D. Zhu, *Acc. Chem. Res.* **2010**, *43*, 529.
[54] Y. Zhang, P. Chen, L. Jiang, W. Hu, M. Liu, *J. Am. Chem. Soc.* **2009**, *131*, 2756.
[55] C. Y. Liu, A. J. Bard, *Acc. Chem. Res.* **1999**, *32*, 235.
[56] K. Thiyagarajan, B. Saravanakumar, S.-J. Kim, *ACS Appl. Mater. Interfaces* **2015**, *7*, 2171.
[57] Z. Wang, X. Zhan, Y. Wang, S. Muhammad, Y. Huang, J. He, *Nanoscale* **2012**, *4*, 2678.
[58] S. Rackauskas, K. Mustonen, T. Jarvinen, M. Mattila, O. Klimova, H. Jiang, O. Tolochko, H. Lipsanen, E. I. Kauppinen, A. G. Nasibulin, *Nanotechnology* **2012**, *23*, 095502.
[59] I. C. Yao, T.-Y. Tseng, P. Lin, *Sens. Actuators, A* **2012**, *178*, 26.
[60] H. T. Zhou, L. Li, H. Y. Chen, Z. Guo, S. J. Jiao, W. J. Sun, *RSC Adv.* **2015**, *5*, 87993.
[61] H. Sirringhaus, *Adv. Mater.* **2005**, *17*, 2411.
[62] Y. Wen, Y. Liu, Y. Guo, G. Yu, W. Hu, *Chem. Rev.* **2011**, *111*, 3358.
[63] A. N. Sokolov, M. E. Roberts, Z. Bao, *Mater. Today* **2009**, *12*, 12.
[64] D. Braga, G. Horowitz, *Adv. Mater.* **2009**, *21*, 1473.
[65] S. Liu, W. M. Wang, A. L. Briseno, S. C. B. Mannsfeld, Z. Bao, *Adv. Mater.* **2009**, *21*, 1217.
[66] P. F. Moonen, I. Yakimets, J. Huskens, *Adv. Mater.* **2012**, *24*, 5526.
[67] M. Plötner, T. Wegener, S. Richter, S. Howitz, W.-J. Fischer, *Synth. Met.* **2004**, *147*, 299.
[68] J.-M. Verilhac, M. Benwadih, A.-L. Seiler, S. Jacob, C. Bory, J. Bablet, M. Heitzman, J. Tallal, L. Barbut, P. Frère, G. Sicard, R. Gwoziecki, I. Chartier, R. Coppard, C. Serbutoviez, *Org. Electron.* **2010**, *11*, 456.
[69] X. Hou, Y. Xia, S. C. Ng, J. Zhang, J. S. Chang, *RSC Adv.* **2014**, *4*, 37687.
[70] B. Kheradmand-Boroujeni, G. C. Schmidt, D. Hoft, R. Shabanpour, C. Perumal, T. Meister, K. Ishida, C. Carta, A. C. Hubler, F. Ellinger, *IEEE Trans. Electron Devices* **2014**, *61*, 1423.
[71] P. Lin, X. Luo, I. M. Hsing, F. Yan, *Adv. Mater.* **2011**, *23*, 4035.
[72] J. Jung, S. J. Kim, K. W. Lee, D. H. Yoon, Y.-g. Kim, H. Y. Kwak, S. R. Dugasani, S. H. Park, H. J. Kim, *Biosens. Bioelectron.* **2014**, *55*, 99.
[73] P. Serre, C. Ternon, V. Stambouli, P. Periwal, T. Baron, *Sens. Actuators, B* **2013**, *182*, 390.
[74] D. Karnaushenko, B. Ibarlucea, S. Lee, G. Lin, L. Baraban, S. Pregl, M. Melzer, D. Makarov, W. M. Weber, T. Mikolajick, O. G. Schmidt, G. Cuniberti, *Adv. Healthcare Mater.* **2015**, *4*, 1517.
[75] Y. H. Kwak, D. S. Choi, Y. N. Kim, H. Kim, D. H. Yoon, S.-S. Ahn, J.-W. Yang, W. S. Yang, S. Seo, *Biosens. Bioelectron.* **2012**, *37*, 82.
[76] D. Lee, T. Cui, *Biosens. Bioelectron.* **2010**, *25*, 2259.
[77] X. You, J. J. Pak, *Sens. Actuators, B* **2014**, *202*, 1357.
[78] T. Minami, T. Sato, T. Minamiki, K. Fukuda, D. Kumaki, S. Tokito, *Biosens. Bioelectron.* **2015**, *74*, 45.
[79] A. K. Diallo, J. Tardy, Z. Q. Zhang, F. Bessueille, N. Jaffrezic-Renault, M. Lemiti, *Appl. Phys. Lett.* **2009**, *94*, 263302.
[80] C.-M. Yang, I. S. Wang, Y.-T. Lin, C.-H. Huang, T.-F. Lu, C.-E. Lue, D. G. Pijanowska, M.-Y. Hua, C.-S. Lai, *Sens. Actuators, B* **2013**, *187*, 274.
[81] M. Qin, Y. Huang, Y. Li, M. Su, B. Chen, H. Sun, P. Yong, C. Ye, F. Li, Y. Song, *Angew. Chem.* **2016**, *128*, 7025.
[82] E. Lindner, V. V. Cosofret, S. Ufer, R. P. Buck, R. P. Kusy, R. B. Ash, H. T. Nagle, *J Chem. Soc. Faraday Trans.* **1993**, *89*, 361.
[83] V. V. Cosofret, M. Erdosy, T. A. Johnson, R. P. Buck, R. B. Ash, M. R. Neuman, *Anal. Chem.* **1995**, *67*, 1647.
[84] R. Koncki, M. Mascini, *Anal. Chim. Acta* **1997**, *351*, 143.
[85] W.-D. Huang, H. Cao, S. Deb, M. Chiao, J. C. Chiao, *Sens. Actuators, A* **2011**, *169*, 1.
[86] C. Hung, V. Landge, U. Tata, S. Young-Sik, S. Rao, T. Shou-Jiang, H. F. Tibbals, S. Spechler, J. Chiao, *IEEE Trans Biomed. Eng.* **2012**, *59*, 3131.
[87] H. J. Chung, M. S. Sulkin, J. S. Kim, C. Goudeseune, H. Y. Chao, J. W. Song, S. Y. Yang, Y. Y. Hsu, R. Ghaffari, I. R. Efimov, *Adv. Healthcare Mater.* **2014**, *3*, 59.
[88] N. Ferrer-Anglada, M. Kaempgen, S. Roth, *Phys. Status Solidi B* **2006**, *243*, 3519.
[89] L. Santos, J. P. Neto, A. Crespo, D. Nunes, N. Costa, I. M. Fonseca, P. Barquinha, L. Pereira, J. Silva, R. Martins, *ACS Appl. Mater. Interfaces* **2014**, *6*, 12226.
[90] A. J. Bandodkar, V. W. Hung, W. Jia, G. Valdés-Ramírez, J. R. Windmiller, A. G. Martinez, J. Ramírez, G. Chan, K. Kerman, J. Wang, *Analyst* **2013**, *138*, 123.
[91] B. Saravanakumar, S. Soyoon, S.-J. Kim, *ACS Appl. Mater. Interfaces* **2014**, *6*, 13716.
[92] R. Rahimi, M. Ochoa, W. Yu, B. Ziaie, in *Solid-State Sensors, Actuators and Microsystems (TRANSDUCERS), Transducers – 18th Int. Conf.*, IEEE, Piscataway, NJ, USA, **2015**.
[93] K. Melzer, V. D. Bhatt, T. Schuster, E. Jaworska, K. Maksymiuk, A. Michalska, P. Lugli, G. Scarpa, *IEEE Sens. J.* **2015**, *15*, 3127.
[94] L. Maiolo, S. Mirabella, F. Maita, A. Alberti, A. Minotti, V. Strano, A. Pecora, Y. Shacham-Diamand, G. Fortunato, *Appl. Phys. Lett.* **2014**, *105*, 093501.
[95] K. Diallo, M. Lemiti, J. Tardy, F. Bessueille, N. Jaffrezic-Renault, *Appl. Phys. Lett.* **2008**, *93*, 183305.
[96] T. Trantidou, M. Tariq, C. M. Terracciano, C. Toumazou, T. Prodromakis, *Sensors* **2014**, *14*, 11629.
[97] J.-S. Chen, J.-C. Chou, Y.-H. Liao, R.-T. Chen, M.-S. Huang, T.-Y. Wu, in *Electron Devices and Solid-State Circuits (EDSSC), IEEE Int. Conf.*, IEEE, Piscataway, NJ, USA, **2015**.
[98] C.-E. Lue, I.-S. Wang, C.-H. Huang, Y.-T. Shiao, H.-C. Wang, C.-M. Yang, S.-H. Hsu, C.-Y. Chang, W. Wang, C.-S. Lai, *Microelectron. Reliab.* **2012**, *52*, 1651.
[99] N. Liu, L. Q. Zhu, P. Feng, C. J. Wan, Y. H. Liu, Y. Shi, Q. Wan, *Sci. Rep.* **2015**, *5*, 18082.
[100] S. Mathison, E. Bakker, *Anal. Chem.* **1998**, *70*, 303.
[101] K. Mitsubayashi, K. Yokoyama, T. Takeuchi, E. Tamiya, I. Karube, *Anal. Chem.* **1993**, *65*, 3586.
[102] A. Michalska, K. Maksymiuk, *Anal. Chim. Acta* **2004**, *523*, 97.
[103] R. L. Portugal Fagury, K. Omuro Lupetti, O. Fatibello-Filho, *Anal. Lett.* **2005**, *38*, 1857.
[104] M. Ganjali, M. Rezapour, M. Pirali-Hamedani, H. Rashedi, *Int. J. Electrochem. Sci.* **2015**, *10*, 6924.
[105] M. Novell, M. Parrilla, G. n. A. Crespo, F. X. Rius, F. J. Andrade, *Anal. Chem.* **2012**, *84*, 4695.
[106] J. Ping, Y. Wang, K. Fan, W. Tang, J. Wu, Y. Ying, *J. Mater. Chem. B* **2013**, *1*, 4781.
[107] K. Dagci, M. Alanyalioglu, *ACS Appl. Mater. Interfaces* **2016**, *8*, 2713.
[108] M. Zhang, A. Halder, C. Hou, J. Ulstrup, Q. Chi, *Bioelectrochemistry* **2016**, *109*, 87.
[109] T. A. Ali, G. G. Mohamed, *J. AOAC Int.* **2015**, *98*, 116.
[110] S. S. Hassan, H. Sayour, S. S. Al-Mehrezi, *Anal. Chim. Acta* **2007**, *581*, 13.
[111] T.-Y. Wu, S.-C. Tseng, J.-C. Chou, Y.-H. Liao, R.-T. Chen, M.-S. Huang, in *Electron Devices and Solid-State Circuits (EDSSC), IEEE Int. Conf.*, IEEE, Piscataway, NJ, USA, **2015**.
[112] R. Thota, V. Ganesh, *Sens. Actuators, B* **2016**, *227*, 169.
[113] W.-Y. Wu, X. Zhong, W. Wang, Q. Miao, J.-J. Zhu, *Electrochem. Commun.* **2010**, *12*, 1600.
[114] A. P. R. de Souza, C. W. Foster, A. V. Kolliopoulos, M. Bertotti, C. E. Banks, *Analyst* **2015**, *140*, 4130.
[115] B. Schazmann, D. Morris, C. Slater, S. Beirne, C. Fay, R. Reuveny, N. Moyna, D. Diamond, *Anal. Methods* **2010**, *2*, 342.
[116] A. Cazalé, W. Sant, F. Ginot, J.-C. Launay, G. Savourey, F. Revol-Cavalier, J.-M. Lagarde, D. Heinry, J. Launay, P. Temple-Boyer, *Sens. Actuators, B* **2016**, *225*, 1.

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[117] E. K. Wujcik, N. J. Blasdel, D. Trowbridge, C. N. Monty, *IEEE Sensors J.* **2013**, *13*, 3430.
[118] T. Guinovart, A. J. Bandodkar, J. R. Windmiller, F. J. Andrade, J. Wang, *Analyst* **2013**, *138*, 7031.
[119] A. J. Bandodkar, D. Molinnus, O. Mirza, T. Guinovart, J. R. Windmiller, G. Valdés-Ramírez, F. J. Andrade, M. J. Schöning, J. Wang, *Biosens. Bioelectron.* **2014**, *54*, 603.
[120] W. Gao, S. Emaminejad, H. Y. Y. Nyein, S. Challa, K. Chen, A. Peck, H. M. Fahad, H. Ota, H. Shiraki, D. Kiriya, *Nature* **2016**, *529*, 509.
[121] T. Guinovart, M. Parrilla, G. A. Crespo, F. X. Rius, F. J. Andrade, *Analyst* **2013**, *138*, 5208.
[122] R. Rahimi, M. Ochoa, B. Ziaie, in *SENSORS, IEEE*, IEEE, Piscataway, NJ, USA **2013**.
[123] T. Guinovart, G. Valdés-Ramírez, J. R. Windmiller, F. J. Andrade, J. Wang, *Electroanalysis* **2014**, *26*, 1345.
[124] G. Matzeu, L. Florea, D. Diamond, *Sens. Actuators, B* **2015**, *211*, 403.
[125] K. Takei, W. Honda, S. Harada, T. Arie, S. Akita, *Adv. Healthcare Mater.* **2015**, *4*, 487.
[126] J. H. An, S. J. Park, O. S. Kwon, J. Bae, J. Jang, *ACS Nano* **2013**, *7*, 10563.
[127] S. Jiang, R. Cheng, R. Ng, Y. Huang, X. Duan, *Nano Res.* **2015**, *8*, 257.
[128] Z. Mousavi, A. Ekholm, J. Bobacka, A. Ivaska, *Electroanalysis* **2009**, *21*, 472.
[129] a) Q. Liu, K. Xiao, L. Wen, H. Lu, Y. Liu, X.-Y. Kong, G. Xie, Z. Zhang, Z. Bo, L. Jiang, *J. Am. Chem. Soc.* **2015**, *137*, 11976; b) Q. Zhai, J. Wang, H. Jiang, Q. Wei, E. Wang, *Talanta* **2015**, *140*, 219.
[130] Y. Huang, F. Li, M. Qin, L. Jiang, Y. Song, *Angew. Chem., Int. Ed.* **2013**, *52*, 7296.
[131] O. Korostynska, M. Ortoneda-Pedrola, A. Mason, A. Al-Shamma'a, *Meas. Sci. Technol.* **2014**, *25*, 065105.
[132] U. Zschieschang, H. Klauk, M. Halik, G. Schmid, C. Dehm, *Adv. Mater.* **2003**, *15*, 1147–1151.
[133] J. T. Mabeck, G. G. Malliaras, *Anal. Bioanal. Chem.* **2006**, *384*, 343.
[134] V. Dua, S. P. Surwade, S. Ammu, S. R. Agnihotra, S. Jain, K. E. Roberts, S. Park, R. S. Ruoff, S. K. Manohar, *Angew. Chem., Int. Ed.* **2010**, *49*, 2154–2157.
[135] B. Crone, A. Dodabalapur, A. Gelperin, L. Torsi, H. E. Katz, A. J. Lovinger, Z. Bao, *Appl. Phys. Lett.* **2001**, *78*, 2229.
[136] Y. Noguchi, T. Sekitani, T. Someya, *Appl. Phys. Lett.* **2006**, *89*, 253507.
[137] M. F. Mabrook, C. Pearson, M. C. Petty, *IEEE Sens. J.* **2006**, *6*, 1435.
[138] J. B. Chang, V. Liu, V. Subramanian, K. Sivula, C. Luscombe, A. Murphy, J. Liu, J. M. J. Fréchet, *J. Appl. Phys.* **2006**, *100*, 014506.
[139] S. Mandal, Y.-Y. Noh, *Semicond. Sci. Technol.* **2015**, *30*, 064003.
[140] F. Liao, C. Chen, V. Subramanian, *Sens. Actuators, B* **2005**, *107*, 849.
[141] X. Xiao, L. Yuan, J. Zhong, T. Ding, Y. Liu, Z. Cai, Y. Rong, H. Han, J. Zhou, Z. L. Wang, *Adv. Mater.* **2011**, *23*, 5440.
[142] N.-K. Chang, C.-C. Su, S.-H. Chang, *Appl. Phys. Lett.* **2008**, *92*, 063501.
[143] S. Park, M. Vosguerichian, Z. Bao, *Nanoscale* **2013**, *5*, 1727.
[144] Y. J. Choi, I. S. Hwang, J. G. Park, K. J. Choi, J. H. Park, J. H. Lee, *Nanotechnology* **2008**, *19*, 095508.
[145] S. Gong, W. Schwalb, Y. Wang, Y. Chen, Y. Tang, J. Si, B. Shirinzadeh, W. Cheng, *Nat. Commun.* **2014**, *5*, 3132.
[146] S. C. Mannsfeld, B. C. Tee, R. M. Stoltenberg, C. V. Chen, S. Barman, B. V. Muir, A. N. Sokolov, C. Reese, Z. Bao, *Nat. Mater.* **2010**, *9*, 859.
[147] D. J. Lipomi, M. Vosgueritchian, B. C. Tee, S. L. Hellstrom, J. A. Lee, C. H. Fox, Z. Bao, *Nat. Nanotechnol.* **2011**, *6*, 788.
[148] T. Someya, T. Sekitani, S. Iba, Y. Kato, H. Kawaguchi, T. Sakurai, *Proc. Natl. Acad. Sci. USA* **2004**, *101*, 9966.
[149] K. Parikh, K. Cattanach, R. Rao, D.-S. Suh, A. Wu, S. K. Manohar, *Sens. Actuators, B* **2006**, *113*, 55.
[150] P.-G. Su, C.-T. Lee, C.-Y. Chou, K.-H. Cheng, Y.-S. Chuang, *Sens. Actuators, B* **2009**, *139*, 488.
[151] Q. He, Z. Zeng, Z. Yin, H. Li, S. Wu, X. Huang, H. Zhang, *Small* **2012**, *8*, 2994.
[152] Y. Kato, S. Iba, R. Teramoto, T. Sekitani, T. Someya, H. Kawaguchi, T. Sakurai, *Appl. Phys. Lett.* **2004**, *84*, 3789.
[153] C. Y. Han, W. M. Tang, C. H. Leung, C. M. Che, P. T. Lai, *Org. Electron.* **2014**, *15*, 2499.
[154] Q.-J. Sun, J. Zhuang, Y. Yan, Y. Zhou, S.-T. Han, L. Zhou, V. A. L. Roy, *Phys. Status Solidi A* **2016**, *213*, 79.
[155] F. Yakuphanoglu, W. A. Farooq, *Synth. Met.* **2011**, *161*, 379.
[156] J. Chen, R. Könenkamp, *Appl. Phys. Lett.* **2003**, *82*, 4782.
[157] W. F. Zhang, Z. B. He, G. D. Yuan, J. S. Jie, L. B. Luo, X. J. Zhang, Z. H. Chen, C. S. Lee, W. J. Zhang, S. T. Lee, *Appl. Phys. Lett.* **2009**, *94*, 123103.
[158] S. Han, W. Jin, D. Zhang, T. Tang, C. Li, X. Liu, Z. Liu, B. Lei, C. Zhou, *Chem. Phys. Lett.* **2004**, *389*, 176.
[159] K. Takei, T. Takahashi, J. C. Ho, H. Ko, A. G. Gillies, P. W. Leu, R. S. Fearing, A. Javey, *Nat. Mater.* **2010**, *9*, 821.
[160] D. Van Der Marel, H. Molegraaf, J. Zaanen, Z. Nussinov, F. Carbone, A. Damascelli, H. Eisaki, M. Greven, P. Kes, M. Li, *Nature* **2003**, *425*, 271.
[161] J. Liu, W. Wu, S. Bai, Y. Qin, *ACS Appl. Mater. Interfaces* **2011**, *3*, 4197.
[162] X. Wang, Y. Gu, Z. Xiong, Z. Cui, T. Zhang, *Adv. Mater.* **2014**, *26*, 1336.
[163] (a) Y. Yuan, G. Giri, A. L. Ayzner, A. P. Zoombelt, S. C. Mannsfeld, J. Chen, D. Nordlund, M. F. Toney, J. Huang, Z. Bao, *Nat. Commun.* **2014**, *5*, 3005; (b) M. Su, F. Li, S. Chen, Z. Huang, M. Qin, W. Li, X. Zhang, Y. Song, *Adv. Mater.* **2016**, *28*, 1369.
[164] M. Zirkl, A. Haase, A. Fian, H. Schön, C. Sommer, G. Jakopic, G. Leising, B. Stadlober, I. Graz, N. Gaar, R. Schwödiauer, S. Bauer-Gogonea, S. Bauer, *Adv. Mater.* **2007**, *19*, 2241.
[165] H.-C. Yuan, J. Shin, G. Qin, L. Sun, P. Bhattacharya, M. G. Lagally, G. K. Celler, Z. Ma, *Appl. Phys. Lett.* **2009**, *94*, 013102.
[166] A. S. G. Reddy, B. B. Narakathu, M. Z. Atashbar, M. Rebros, E. Rebrosova, M. K. Joyce, *Procedia Eng.* **2011**, *25*, 120.
[167] S. Khan, S. Tinku, L. Lorenzelli, R. S. Dahiya, *IEEE Sensors J.* **2015**, *15*, 3146.
[168] W.-Y. Chang, T.-H. Fang, H.-J. Lin, Y.-T. Shen, Y.-C. Lin, *J Disp. Technol.* **2009**, *5*, 178.
[169] *SENSORS MARKETS 2016: Worldwide Analyses and Forecasts of Sensors Markets until 2016*, INTECHNO CONSULTING, Basel, Switzerland **2012**.
[170] *Wearable Sensors 2015–2025: Market Forecasts, Technologies, Players*, IDTechEx, Cambridge, UK **2015**.
[171] C. Guillaume, *Printed and Flexible Sensors 2015–2025: Technologies, Players, Forecasts*, IDTechEx, Cambridge, UK **2015**.



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